

Chapter 03 :- Functions of Several variables.

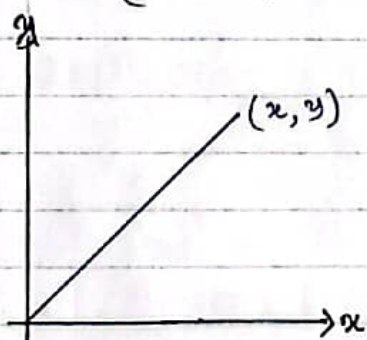
$\mathbb{R}$: The Set of real Numbers.

$$\mathbb{R}^n = \left\{ (x_1, x_2, \dots, x_n) \mid x_i \in \mathbb{R} \text{ for all } i = 1, 2, \dots, n \right\}$$

A typical element $x = (x_1, x_2, \dots, x_n)$ in $\mathbb{R}^n$ is called an ordered n -tuple.

when $n=2$,

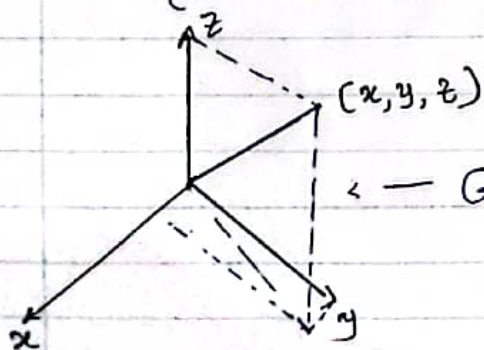
$$\begin{aligned} \mathbb{R}^2 &= \left\{ (x_1, x_2) \mid x_1, x_2 \in \mathbb{R} \right\} \\ &= \{x, y \mid x, y \in \mathbb{R}\}. \end{aligned}$$



Geometrical representation of $\mathbb{R}^2$

when $n=3$,

$$\begin{aligned} \mathbb{R}^3 &= \left\{ (x_1, x_2, x_3) \mid x_1, x_2, x_3 \in \mathbb{R} \right\} \\ &= \{x, y, z \mid x, y, z \in \mathbb{R}\}. \end{aligned}$$

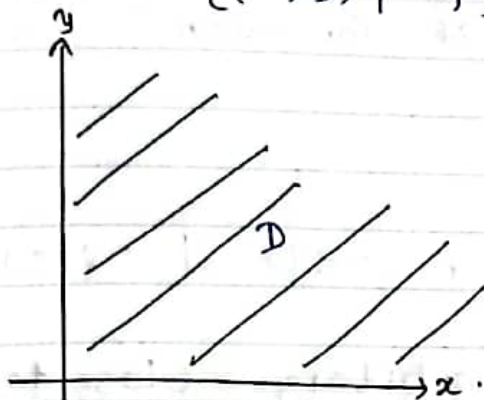


Geometrical Representation of $\mathbb{R}^3$

In this Chapter we shall study functions $f: D \rightarrow \mathbb{R}$, where $D \subseteq \mathbb{R}^2$ or $D \subseteq \mathbb{R}^3$.

Example 01;

Let $D = \{(x, y) \mid x, y > 0\}$.



Define $f: D \rightarrow \mathbb{R}$ by $f(x, y) = x, y$

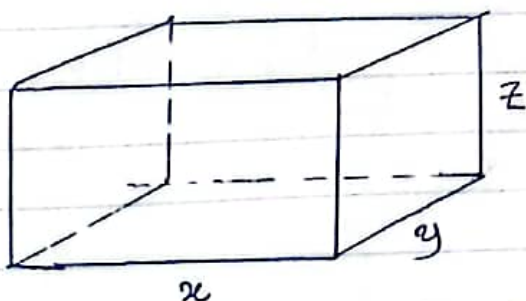
This is a function of two variables. It gives the area of the unbounded region D.

Example 02:

Let $D = \{(x, y, z) \mid x, y, z > 0\}$.

Define $U: D \rightarrow \mathbb{R}$ by $U(x, y, z) = xyz$

This is a function of three variables. It gives the volume of the unbounded cube.



The Limit of a function;

Note that;

when $n=1$, for a function $f: \mathbb{R} \rightarrow \mathbb{R}$,
we have

$$\lim_{x \rightarrow a} f(x) = l \in \mathbb{R} \text{ if for each}$$

$$\epsilon > 0 \quad \exists \quad \delta(\epsilon) > 0$$

Such that

$$|f(x) - l| < \epsilon \quad \text{whenever} \quad 0 < |x - a| < \delta(\epsilon)$$

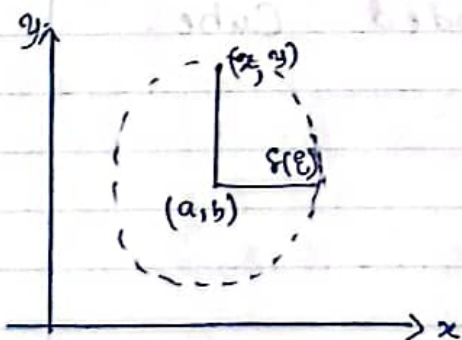
i.e; $f(x)$ can be made arbitrary close to l by
making x sufficiently close to a but
not $x = a$.

Definition :-

Let D is a Subset of a $\mathbb{R}^2$ ($D \subseteq \mathbb{R}^2$) and
 $f: D \rightarrow \mathbb{R}$

Then limit $f(x, y) = l$.
 $(x, y) \rightarrow (a, b)$

if for each $\epsilon > 0 \quad \exists \quad \delta(\epsilon) > 0$ Such that
 $|f(x, y) - l| < \epsilon$ whenever $0 < \sqrt{(x-a)^2 + (y-b)^2} < \delta(\epsilon)$



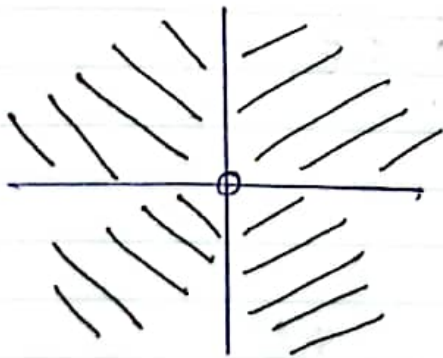
Example 03:

Prove that $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 y}{x^2 + y^2} = 0$.

Solution :-

Let $f(x,y) = \frac{x^2 y}{x^2 + y^2}$ Also Let $l = 0$.

The domain of $D(f) = \{(x,y) \mid (x,y) \neq (0,0)\}$
 $= \mathbb{R}^2 - \{(0,0)\}$



Take any $\epsilon > 0$

Consider

$|f(x,y) - l|$, So that

$$f(x,y) = \frac{x^2 y}{x^2 + y^2} \text{ and } l = 0$$

$$|f(x,y) - l| = \left| \frac{x^2 y}{x^2 + y^2} - 0 \right|$$

$$= \left| \frac{x^2 y}{x^2 + y^2} \right|$$

$$= x^2 \left| \frac{y}{x^2 + y^2} \right| = \frac{x^2}{x^2 + y^2} |y|$$

$$\sqrt{x^2} = |x|$$

$$\sqrt{(-2)^2} = |-2| = 2$$

$$\sqrt{9} = 3$$

$$(3^2)^2 = \pm 3$$

$$x^2 \leq x^2 + y^2$$

$$\frac{x^2}{x^2 + y^2} \leq 1$$

$$\sqrt{y^2} \leq \sqrt{x^2 + y^2}$$

$$|y| \leq \sqrt{x^2 + y^2}$$

$$|f(x, y) - l| = \frac{x^2}{x^2 + y^2} |y|$$

$$\leq |y| \text{ Since } \frac{x^2}{x^2 + y^2} \leq 1$$

$$\leq \sqrt{x^2 + y^2}$$

$$\leq \epsilon \text{ whenever, } 0 < \sqrt{x^2 + y^2} < \epsilon$$

Then we have

$$|f(x, y) - l| < \epsilon \text{ whenever}$$

$\nearrow$

$$0 < \sqrt{(x-0)^2 + (y-0)^2} < \delta(\epsilon)$$

$$\therefore \lim_{(x, y) \rightarrow (0, 0)} f(x, y) = 0$$

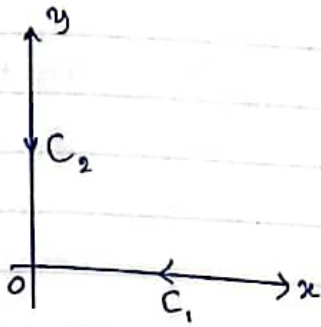
(Homework)

$$\text{Prove that } \lim_{(x, y) \rightarrow (0, 0)} \frac{3xy^2}{x^2 + y^2} = 0$$

Example 04:-

Show that $\lim_{(x, y) \rightarrow (0, 0)} \frac{x^2 - y^2}{x^2 + y^2}$ does not exist.

$$D = \mathbb{R}^2 - \{(0, 0)\} \leftarrow \text{Domain of } f(x, y) = \frac{x^2 - y^2}{x^2 + y^2}$$



Along $C_1 : y=0$ and $x>0$

$$f(x, y) = \frac{x^2 - 0}{x^2 + 0^2} = \frac{x^2}{x^2} = 1$$

$$\lim_{(x, y) \rightarrow (0, 0)} f(x, y) = \lim_{x \rightarrow 0} f(x, y)$$

$$= \lim_{x \rightarrow 0} 1$$

$$= 1 \quad \text{--- (1)}$$

Along $C_2 : x=0$ and $y>0$.

$$f(x, y) = \frac{x^2 - y^2}{x^2 + y^2} = \frac{0 - y^2}{0^2 + y^2} = \frac{-y^2}{y^2} = -1$$

Now, $\lim_{(x, y) \rightarrow (0, 0)} f(x, y) = \lim_{y \rightarrow 0} f(x, y)$

$$= \lim_{y \rightarrow 0} -1$$

$$= -1 \quad \text{--- (2)}$$

By (1) and (2), we conclude that

$$\lim_{(x, y) \rightarrow (0, 0)} f(x, y) = \text{DNE.}$$

(Homework) 1) Show that $\lim_{(x, y) \rightarrow (0, 0)} \frac{xy}{x^2 + y^2} \text{ DNE.}$

2) Show that $\lim_{(x, y) \rightarrow (0, 0)} \frac{xy^2}{x^2 + y^4} \text{ DNE.}$

Note:- The definition of $\lim_{(x, y) \rightarrow (a, b)} f(x, y) = l$

refers only to the distance between (x, y) and (a, b) . It does not refer to the direction of approach.

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If $\lim_{(x,y) \rightarrow (a,b)} f(x,y)$ exists, No matter, how (x,y) approaches (a,b)



Definition:-

Let $D \subseteq \mathbb{R}^2$ and $f: D \rightarrow \mathbb{R}$

f is said to be Continuous at (a,b) in D if

$$\lim_{(x,y) \rightarrow (a,b)} f(x,y) = f(a,b)$$

i.e:-

- 1) $f(a,b)$ must be defined
- 2) $\lim_{(x,y) \rightarrow (a,b)} f(x,y)$ must exist.
- 3) $\lim_{(x,y) \rightarrow (a,b)} f(x,y) = f(a,b)$

f is cts at (a,b)

Example 5.

- a) Let $f(x,y) = \frac{x^2 y}{x^2 + y^2}$ IS f continuous at $(0,0)$?

No! Since $f(0,0)$ is not define.

b) Let $f(x, y) = \begin{cases} \frac{x^2 y}{x^2 + y^2} & \text{if } (x, y) \neq (0, 0) \\ 0 & \text{if } (x, y) = (0, 0) \end{cases}$

Is f Continuous at $(0, 0)$?

Clearly, $f(0, 0)$ is defined,
we have already seen in Example that
 $\lim_{(x, y) \rightarrow (0, 0)} f(x, y) = 0$

But $\lim_{(x, y) \rightarrow (0, 0)} f(x, y) \neq f(0, 0)$

$\therefore f$ is not Continuous at $(0, 0)$.

Definition:-

Let $D \subseteq \mathbb{R}^2$ and $f: D \rightarrow \mathbb{R}$ be a function.
 f is said to be Continuous on D if f is
Continuous at each point in D .

Example 06:-
Let $g(x, y) = \begin{cases} \frac{x^2 y}{x^2 + y^2} & \text{if } (x, y) \neq (0, 0) \\ 0 & \text{if } (x, y) = (0, 0) \end{cases}$

Since $\lim_{(x, y) \rightarrow (0, 0)} g(x, y) = 0 = f(0, 0)$

So, g is continuous at $(0, 0)$

In fact, f is Cts on $\mathbb{R}^2$ (Homework)

Partial Differentiation.

Let $f(x, y) = x^2 + y^2$

$\frac{\partial f}{\partial x} = 2x$ $\leftarrow$ Regarding y as a constant and differentiating $f(x, y)$ w.r.t. x .

$\frac{\partial f}{\partial y} = 2y$ $\leftarrow$ Regarding x as a constant and differentiating $f(x, y)$ w.r.t. y .

Let $f(x, y, z) = xy + yz + z^2$

$\frac{\partial f}{\partial x} = y$

$\frac{\partial f}{\partial z} = z + 2z$

$\frac{\partial f}{\partial y} = x + z$

Let $f(x, y) = e^{x^2+y^2} \sin(x+2y)$

$$\begin{aligned} \frac{\partial f}{\partial x} &= e^{x^2+y^2} (\cos(x+2y) \times 1) + \sin(x+2y) e^{x^2+y^2} \times 2x \\ &= e^{x^2+y^2} [2x \sin(x+2y) + \cos(x+2y)] \end{aligned}$$

$$\begin{aligned} \frac{\partial f}{\partial y} &= e^{x^2+y^2} \times 2y \sin(x+2y) + e^{x^2+y^2} \cos(x+2y) \times 2 \\ &= 2e^{x^2+y^2} [y \sin(x+2y) + \cos(x+2y)] \end{aligned}$$

Let $f(x, y) = x^3 + 3x^2y + xy^7$.

$$\frac{\partial f}{\partial x} = 3x^2 + 6xy + y^7$$

$$\frac{\partial}{\partial x} \left(\frac{\partial f}{\partial x} \right) = 6x + 6y = 6(x+y)$$

$$\frac{\partial f}{\partial y} = 3x^2 + 7xy^6$$

$$= \frac{\partial^2 f}{\partial x^2}$$

$$\frac{\partial^2 f}{\partial y \partial x} = \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial x} \right) = 6x + 7y^6$$

Definition:-

Let $D \subseteq \mathbb{R}^n$ be an open set and Let $a = (a_1, a_2, \dots, a_n)$ in D .

Let $f: D \rightarrow \mathbb{R}$.

The Partial derivative of f w.r.t the i^{th} Co-ordinate at a is defined to be

$$\lim_{h \rightarrow 0} \left[\frac{f(a_1, a_2, \dots, a_i + h, a_{i+1}, \dots, a_n) - f(a_1, a_2, \dots, a_n)}{h} \right]$$

Provide that the limit exists.

This is denoted by $\frac{\partial f(a)}{\partial x_i}$

Other notation:

(i) $D_i(a)$

(ii) $f_{x_i}(a)$

when $n=1$

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

when $n=2$

$$a = (x_0, y_0) :$$

$$\frac{\partial f}{\partial x} = \lim_{h \rightarrow 0} \frac{f(x_0 + h, y_0) - f(x_0, y_0)}{h} = f_x(x_0, y_0)$$

$$\frac{\partial f}{\partial y} = \lim_{k \rightarrow 0} \frac{f(x_0, y_0 + k) - f(x_0, y_0)}{k} = f_y(x_0, y_0)$$

Note:- Let $z = f(x, y)$

Let Δx and Δy be increments w.r.t x and y respectively.

$$\text{Then, } \Delta z = f(x + \Delta x, y + \Delta y) - f(x, y)$$

The differential of f (or z) denoted by df (or dz) and is defined by.

$$\boxed{dz = \frac{\partial z}{\partial x} dx + \frac{\partial z}{\partial y} dy} \leftarrow \text{Total differential of } f \text{ (or } z \text{)}$$

Example:-

$$\text{Let } f = \begin{cases} \frac{xy}{x^2 + y^2} & \text{if } (x, y) \neq (0, 0) \\ 0 & \text{if } (x, y) = (0, 0) \end{cases}$$

we have already seen that f is not continuous at $(0, 0)$

when $(x, y) \neq (0, 0)$

$$f(x, y) = \frac{xy}{x^2 + y^2}$$

$$\frac{\partial f}{\partial x} = \frac{(x^2+y^2)y - xy \times 2x}{(x^2+y^2)^2} = \frac{y^3 - x^2y}{(x^2+y^2)^2} = \frac{y(y^2-x^2)}{(x^2+y^2)^2}$$

$$\frac{\partial f}{\partial y} = \frac{(x^2+y^2)x - xy \times 2y}{(x^2+y^2)^2} = \frac{x^3 - xy^2}{(x^2+y^2)^2} = \frac{x(x^2-y^2)}{(x^2+y^2)^2}$$

Do $\frac{\partial f}{\partial x} \Big|_{(0,0)}$ and $\frac{\partial f}{\partial y} \Big|_{(0,0)}$ exist?

By definition

$$\begin{aligned} \frac{\partial f}{\partial x} &= \lim_{h \rightarrow 0} \frac{f(0+h, 0) - f(0, 0)}{h} = \lim_{h \rightarrow 0} \frac{f(h, 0) - f(0, 0)}{h} \\ &= \lim_{h \rightarrow 0} \frac{0 - 0}{h} = 0. \end{aligned}$$

¶

Since $f(h, 0) = 0$ and $f(0, 0) = 0$.

$$\text{Also, } f_y(0, 0) = \lim_{k \rightarrow 0} \frac{f(0, 0+k) - f(0, 0)}{k}$$

$$= \lim_{k \rightarrow 0} \frac{f(0, k) - f(0, 0)}{k}$$

$$= \lim_{k \rightarrow 0} \frac{0 - 0}{k}$$

Since $f(0, k) = 0$ and $f(0, 0) = 0$

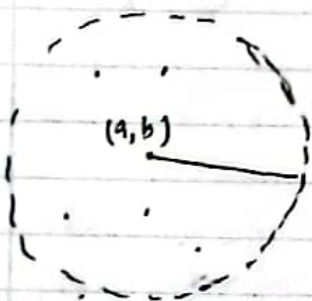
$$= 0$$

Both $f_x(0, 0)$ and $f_y(0, 0)$ exist at $(0, 0)$

* But f is not continuous at $(0, 0)$. that is the existence of the first partial derivatives at a point does not imply continuity at a point.

Differentiability of functions of two variables.

Let f be a function defined in some neighbourhood of (a, b) .



Suppose that $f_x(a, b)$ and $f_y(a, b)$ exist. Then f is said to be differentiable at (a, b) if this limit.

$$\lim_{(h, k) \rightarrow (0, 0)} \frac{f(a+h, b+k) - f(a, b) - hf_x(a, b) - kf_y(a, b)}{\sqrt{h^2 + k^2}} = 0.$$

example:-

Let $f(x, y) = 2x^2y$ use the definition to show that f is differentiable at $(2, 3)$.

Let $a = 2$, $b = 3$.

$$\text{Let } R(h, k) = \frac{f(a+h, b+k) - f(a, b) - hf_x(a, b) - kf_y(a, b)}{\sqrt{h^2 + k^2}}$$

We must show that $\lim_{(h, k) \rightarrow (0, 0)} R(h, k) = 0$.

$$f(x, y) = 2x^2y \Rightarrow f_x(x, y) = 4xy \text{ and } f_y(x, y) = 2x^2$$
$$f(2, 3) = 24 \text{ and } f_x(2, 3) = 24 \text{ and } f_y(2, 3) = 8.$$

$$f(2+h, 3+k) = 2(2+h)^2(3+k)$$

$$\begin{aligned}
 R(h,k) &= \frac{2(2+h)^2(3+k) - 24 - 24h - 8k}{\sqrt{h^2+k^2}} \\
 &= \frac{2(4+4h+h^2)(3+k) - 24 - 24h - 8k}{\sqrt{h^2+k^2}} \\
 &= \frac{8hk + 2h^2k + 6h^2}{\sqrt{h^2+k^2}} \\
 &= \frac{8hk}{\sqrt{h^2+k^2}} + \frac{2h^2k}{\sqrt{h^2+k^2}} + \frac{6h^2}{\sqrt{h^2+k^2}} \\
 &= R_1(h,k) + R_2(h,k) + R_3(h,k) \quad \text{--- ①}
 \end{aligned}$$

$$\lim_{(h,k) \rightarrow (0,0)} R(h,k) = \lim_{(h,k) \rightarrow (0,0)} R_1(h,k) + \lim_{(h,k) \rightarrow (0,0)} R_2(h,k) + \lim_{(h,k) \rightarrow (0,0)} R_3(h,k)$$

Take any $\epsilon > 0$.

$$\left| \frac{hk}{\sqrt{h^2+k^2}} - 0 \right| = \frac{hk}{\sqrt{h^2+k^2}} \leq |h| \leq \sqrt{h^2+k^2}$$

$$\begin{aligned}
 &\leq |h| \\
 &\leq \sqrt{h^2+k^2} \\
 &< \epsilon
 \end{aligned}$$

whenever $0 < \sqrt{h^2+k^2} < \delta = \delta(\epsilon)$

$$\therefore \text{By def}^n: \lim_{(h,k) \rightarrow (0,0)} \frac{hk}{\sqrt{h^2+k^2}} = 0$$

$$\therefore R_1(h,k) = \frac{8hk}{\sqrt{h^2+k^2}}$$

$$\begin{aligned}
 \lim_{(h,k) \rightarrow (0,0)} R_1(h,k) &= \lim_{(h,k) \rightarrow (0,0)} \frac{8hk}{\sqrt{h^2+k^2}} = 8 \lim_{(h,k) \rightarrow (0,0)} \frac{hk}{\sqrt{h^2+k^2}} \\
 &= 0
 \end{aligned}$$

Likewise, we can show that,
 $\lim_{(h,k) \rightarrow (0,0)} R_2(h,k) = 0$ and $\lim_{(h,k) \rightarrow (0,0)} R_3(h,k) = 0$.

$\therefore$ By ①, we have $\lim_{(h,k) \rightarrow (0,0)} R(h,k) = 0$.

$\therefore f$ is differentiable at $(2,3)$ //

* Theorem:-

If a function $f = f(x,y)$ is differentiable at (a,b) then f is continuous at (a,b)

Second order derivatives:-

Let $f = f(x,y)$ Suppose that $f_x(x,y)$ and $f_y(x,y)$ exist.

$$f_x = \frac{\partial f}{\partial x} \Rightarrow \frac{\partial^2 f}{\partial x^2} = \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial x} \right) = \frac{\partial}{\partial x} (f_x) = f_{xx}.$$

$$f_y = \frac{\partial f}{\partial y} \Rightarrow \frac{\partial^2 f}{\partial y^2} = \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial y} \right) = \frac{\partial}{\partial y} (f_y) = f_{yy}.$$

$$\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial y} \right) = \frac{\partial}{\partial x} (f_y) = f_{xy}$$

$$\frac{\partial^2 f}{\partial y \partial x} = \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial x} \right) = \frac{\partial}{\partial y} (f_x) = f_{yx}.$$

Note:- In general that,

$$f_{xy} \neq f_{yx}$$

Example :-

$$\text{Let } f(x, y) = x^3 y + e^{xy^2}$$
$$f_x = 3x^2 y + e^{xy^2} \times y^2 = 3x^2 y + y^2 e^{xy^2}$$

$$\frac{\partial^2 f}{\partial x^2} = \frac{\partial}{\partial x} (f_x) = 6xy + y^2 e^{xy^2}$$

$$\frac{\partial^2 f}{\partial y \partial x} = \frac{\partial f}{\partial y} \left(\frac{\partial f}{\partial x} \right) = \frac{\partial f}{\partial y} (f_x) = \frac{\partial}{\partial y} [3x^2 y + y^2 e^{xy^2}]$$
$$= 3x^2 + y^2 e^{xy^2} \times 2yx + e^{xy^2} \times 2y$$
$$= 3x^2 + 2y^{xy^2} [y^2 x + 1] \quad \text{--- (1)}$$

$$f_y = \frac{\partial f}{\partial y} = x^3 + e^{xy^2} \times 2yx$$

$$\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial y} \right) = \frac{\partial}{\partial x} (x^3 + 2yx e^{xy^2})$$
$$= 3x^2 + 2xy^2 e^{xy^2} + e^{xy^2} \times 2y$$
$$= 3x^2 + 2ye^{xy^2} [xy^2 + 1] \quad \text{--- (2)}$$

Homework :-

* Find an example for a function $f = f(x, y)$ such that, $\frac{\partial^2 f}{\partial x \partial y} \neq \frac{\partial^2 f}{\partial y \partial x}$.

$$f = f(x, y)$$

Theorem :-

If f_x and f_y are both differentiable at a point (a, b) of the domain of a function f , then,

$$f_{xy}(a, b) = f_{yx}(a, b)$$

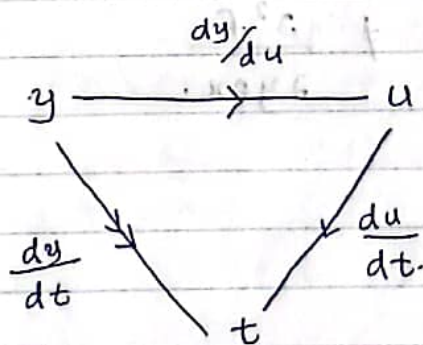
Theorem :-

If f_y exists in a neighbourhood of a point (a, b) of the domain of the definition of a function f and f_{yx} is continuous at (a, b) , then $f_{xy}(a, b)$ exists and $f_{xy}(a, b) = f_{yx}(a, b)$.

— Chain rule —

If $y = f(u)$ is a differentiable function of u and $u = g(t)$, then.

$$\frac{dy}{dt} = \frac{dy}{du} \times \frac{du}{dt}$$

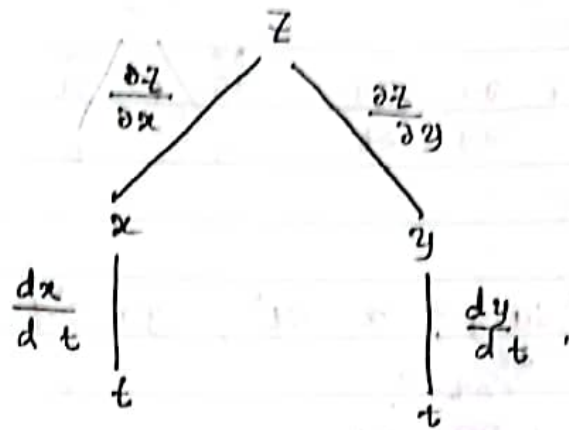


example :-

Let $y = e^{u^2}$ and $u = t^2 + 2t$

$$\begin{aligned} \frac{dy}{dt} &= \frac{dy}{du} \times \frac{du}{dt} = (e^{u^2} \times 2u) (2t + 2) \\ &= 4ute^{u^2} + 4ue^{u^2} \\ &= 4ue^{u^2}(t + 1) \\ &= 4(t^2 + 2t)e^{(t^2 + 2t)^2}(t + 1) // \end{aligned}$$

Let $z = f(x, y)$ be a differentiable function of x and y . Suppose that $x = g(t)$ and $y = h(t)$ are both differentiable functions of t then z is a differentiable function of t and

$$\frac{dz}{dt} = \frac{\partial z}{\partial x} \frac{dx}{dt} + \frac{\partial z}{\partial y} \frac{dy}{dt}.$$


$$z = z(t)$$

$$\frac{dz}{dt} = \frac{\partial z}{\partial x} \frac{dx}{dt} + \frac{\partial z}{\partial y} \frac{dy}{dt}.$$

Example :-

Let $z = e^x \sin y$, $x = t^2$ and $y = \sin t$ find $\frac{dz}{dt}$ by chain rule.

$$\frac{dz}{dt} = \frac{\partial z}{\partial x} \frac{dx}{dt} + \frac{\partial z}{\partial y} \frac{dy}{dt}$$

$$= \frac{\partial}{\partial x} [e^x \sin y] \frac{d}{dt} (t^2) + \frac{\partial}{\partial y} [e^x \sin y] \frac{d}{dt} (\sin t)$$

$$= e^x \sin y \times 2t + e^x \cos y \cos t.$$

$$= e^{t^2} \sin(\sin t) 2t + e^{t^2} \cos(\sin t) \cos t$$

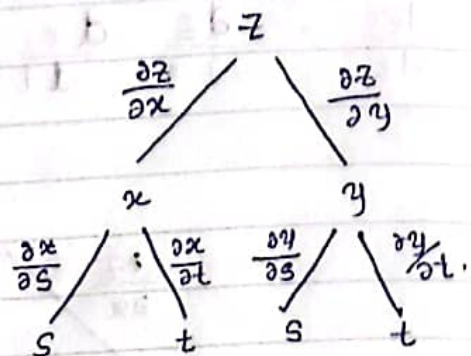
$$= e^{t^2} [2t \sin(\sin t) + \cos t \cos(\sin t)]$$

Let $z = f(x, y)$ be a differentiable function of x and y . Suppose that $x = g(s, t)$ and $y = h(s, t)$ where g_s, g_t, h_s and h_t all exist.

Then,

$$\frac{\partial z}{\partial s} = \frac{\partial z}{\partial x} \frac{\partial x}{\partial s} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial s}$$

$$\frac{\partial z}{\partial t} = \frac{\partial z}{\partial x} \frac{\partial x}{\partial t} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial t}$$



Example:-

Let $z = e^x \sin y$, $x = st^2$, $y = s^3 t$.
find $\frac{\partial z}{\partial s}$ and $\frac{\partial z}{\partial t}$?

$$\frac{\partial z}{\partial s} = \frac{\partial z}{\partial x} \frac{\partial x}{\partial s} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial s}$$

$$= \frac{\partial}{\partial x} (e^x \sin y) \frac{\partial}{\partial s} (st^2) + \frac{\partial}{\partial y} (e^x \sin y) \frac{\partial}{\partial s} (s^3 t)$$

$$= e^x \sin y \times t^2 + e^x \cos y \times 3s^2 t$$

$$= e^{st^2} [t \sin y + 3s^2 \cos y] = te^{st^2} [t \sin(s^3 t) + 3s^2 \cos(s^3 t)]$$

$$\frac{\partial z}{\partial t} = \frac{\partial z}{\partial x} \frac{\partial x}{\partial t} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial t}$$

$$= e^x \sin y \times 2ts + e^x \cos y \times s^3$$

$$= se^{st^2} [2t \sin y + s^2 \cos y]$$

$$= se^{st^2} [2t \sin(s^3 t) + s^2 \cos(s^3 t)]$$

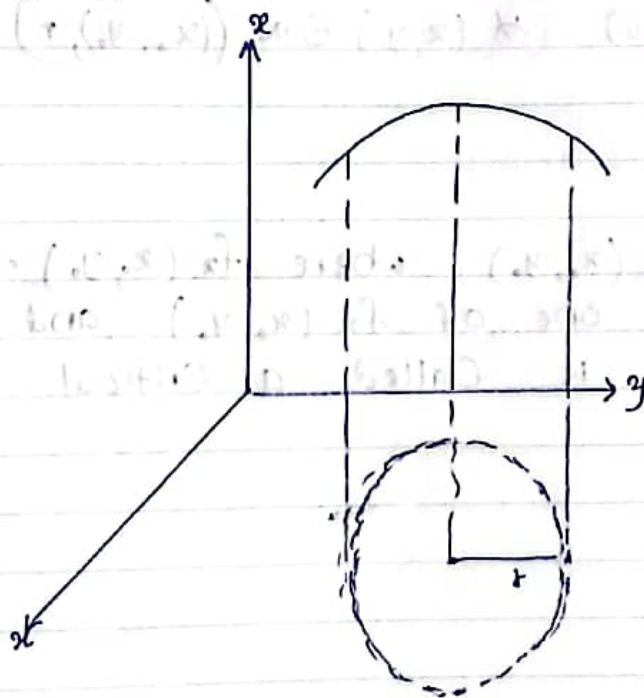
Homework :-

Let $z = f(x, y)$, $x = r^2 + s^2$ and $y = 2rs$ find $\frac{\partial z}{\partial r}$, $\frac{\partial z}{\partial s}$, $\frac{\partial^2 z}{\partial s^2}$, $\frac{\partial^2 z}{\partial s \partial r}$ and $\frac{\partial^2 z}{\partial r^2}$

assuming that,
 $z_{xy} = z_{yx}$.

* Extreme Values of functions of two variables

Let f be a real valued function of two variables define in a neighbourhood of a point (x_0, y_0) the function is said to have a local maximum at (x_0, y_0) if $f(x, y) \leq f(x_0, y_0)$ for all points (x, y) sufficiently closed to (x_0, y_0) .



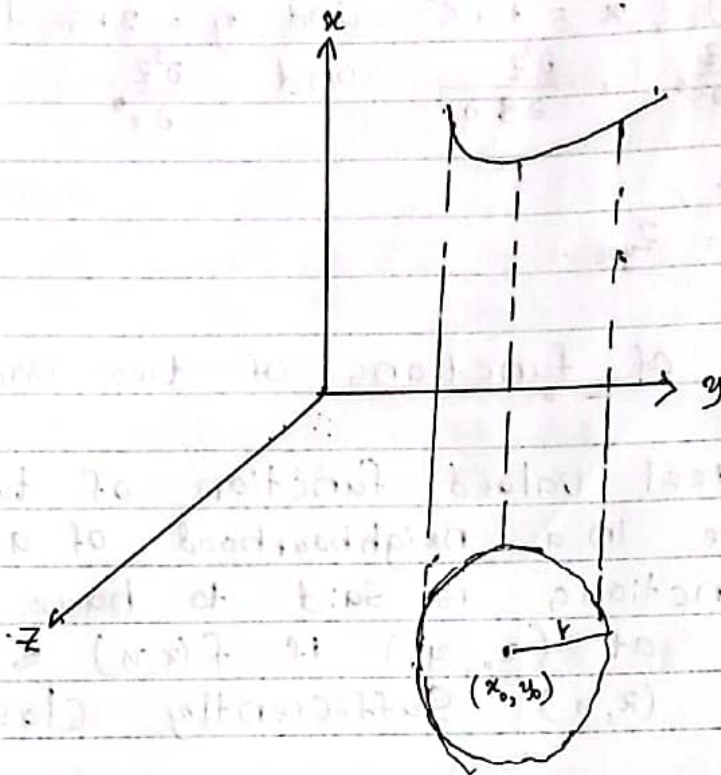
$$N(x_0, y_0, r) = \left\{ (x, y) \in \mathbb{R}^2 \mid 0 < \sqrt{(x-x_0)^2 + (y-y_0)^2} < r \right\}$$

f has a local maximum at (x_0, y_0) if $\exists \delta > 0$

such that,

$$f(x, y) \leq f(x_0, y_0) \quad \forall \quad x, y \in N(x_0, y_0, r)$$

A Local minimum is defined in a similar way.



$$f(x_0, y_0) \leq f(x, y) \quad \forall (x, y) \in N((x_0, y_0), r)$$

Definition :-

A point (x_0, y_0) where $f_x(x_0, y_0) = 0 = f_y(x_0, y_0)$ or at least one of $f_x(x_0, y_0)$ and $f_y(x_0, y_0)$ is undefined is called a Critical point of f .

$$** f_{\min} = f\left(\frac{1}{7}, \frac{3}{14}, -\frac{1}{14}\right)$$

$$= \left(\frac{1}{7}\right)^2 + \left(\frac{3}{14}\right)^2 + \left(-\frac{1}{14}\right)^2$$

$$= \frac{1}{49} + \frac{9}{196} + \frac{1}{196}$$

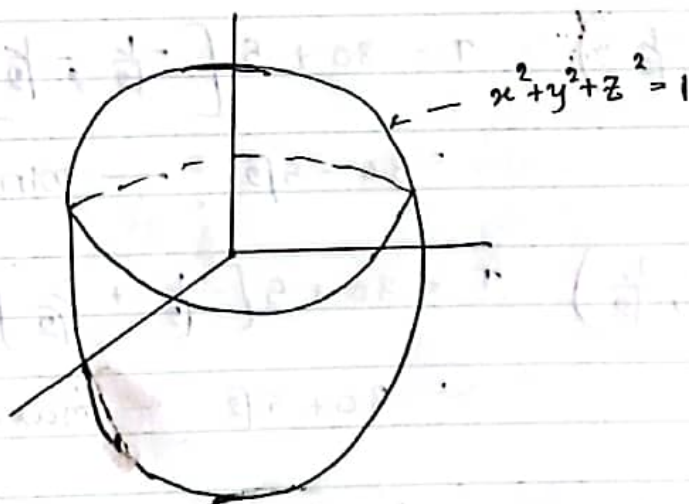
$$= \frac{4 + 9 + 1}{196}$$

$$= \frac{14}{196}$$

$$= \frac{1}{14} > 0$$

Ex or) Suppose that the temperature at a point (x, y, z) on the unit sphere $S: x^2 + y^2 + z^2 = 1$ is given by $T(x, y, z) = 30 + 5(x + z)$.

Find the extremum value of T .



$$\text{Let } F = 30 + 5(x + z) + \lambda(x^2 + y^2 + z^2 - 1)$$

where $\lambda \neq 0$.

Then we have

$$f_x = 5 + 2\lambda x$$

$$f_y = 2\lambda y$$

$$f_z = 5 + 2\lambda z$$

$$f_\lambda = x^2 + y^2 + z^2 - 1$$

No. _____ Date: ____/____/____
Now by Lagrange's multiplier at the Stream values of T , we have

$$5 + 2\lambda x = 0 \quad \text{--- (1)}$$

$$2\lambda y = 0 \quad \text{--- (2)}$$

$$5 + 2\lambda z = 0 \quad \text{--- (3)}$$

$$x^2 + y^2 + z^2 = 1 \quad \text{--- (4)}$$

$$x = \frac{-5}{2\lambda} \quad (\lambda \neq 0) \quad y = 0 \quad (\lambda \neq 0) \quad z = \frac{-5}{2\lambda} \quad (\lambda \neq 0)$$

To find λ , Substitute x, y , and z into (4)

$$\text{So, } \left(\frac{-5}{2\lambda}\right)^2 + 0^2 + \left(\frac{-5}{2\lambda}\right)^2 = 1$$

$$\lambda = \pm \frac{5}{\sqrt{2}}$$

$$\text{So, } x \text{ is } \pm \frac{1}{\sqrt{2}}, \quad y = 0 \text{ and } z = \pm \frac{1}{\sqrt{2}}$$

extreme point are

$$\left(\frac{-1}{\sqrt{2}}, 0, \frac{-1}{\sqrt{2}}\right) \text{ and }$$

$$\left(\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}}\right)$$

$$\text{at } \left(\frac{-1}{\sqrt{2}}, 0, \frac{-1}{\sqrt{2}}\right) \quad T = 30 + 5 \left[\frac{-1}{\sqrt{2}} + \frac{1}{\sqrt{2}} \right]$$

$$= 30 - 5\sqrt{2} \quad \text{--- minimum} //$$

$$\text{at the } \left(\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}}\right) \quad T = 30 + 5 \left[\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} \right]$$

$$= 30 + 5\sqrt{2} \quad \text{--- maximum} //$$

Homework :-

Find the maximum and minimum values of $x^2 + y^2 + z^2$ Subject to the conditions,

$$\frac{x^2}{4} + \frac{y^2}{5} + \frac{z^2}{25} = 1 \quad \text{and}$$

$$z = x + y$$

$$F = x, y, + x, y, z$$

Defⁿ:-

A function $z = f(x, y)$ has a Saddle Point at (x_0, y_0) if (x_0, y_0) is a Critical point of f and for each $\epsilon > 0$ there exist points.

(x_1, y_1) and (x_2, y_2) in $N((x_0, y_0), \epsilon)$ with $f(x_1, y_1) < f(x_0, y_0)$ and $f(x_2, y_2) > f(x_0, y_0)$

Ex 01:-

Let $f(x, y) = y^2 - x^2$

Then we have $f_x = \frac{\partial f}{\partial x} = -2x$ and $f_y = \frac{\partial f}{\partial y} = 2y$.

The Critical points of f are given by $\frac{\partial f}{\partial x} = 0$ and $\frac{\partial f}{\partial y} = 0$.

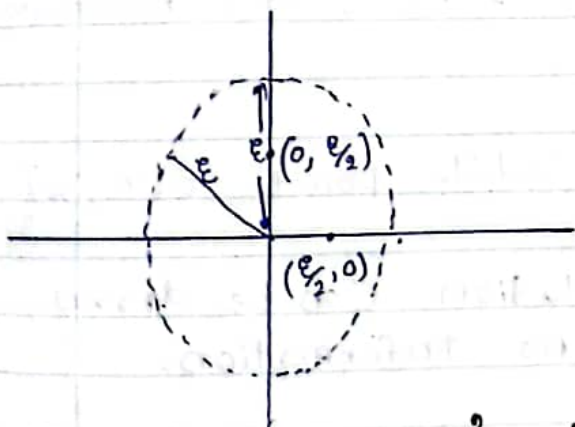
That is $-2x = 0$ and $2y = 0$

So, we have $x=0$ and $y=0$.

Therefore $(0,0)$ is a critical point of f .

Is $(0,0)$ a Saddle point?

Take any $\epsilon > 0$ (Consider nbhd $N((0,0), \epsilon)$)



$f(0,0) = 0$ Also, $(\epsilon/2, 0)$ and $(0, \epsilon/2)$ are in $N((0,0), \epsilon)$.

$$f(0, \epsilon/2) = (\epsilon/2)^2 - 0^2 = (\epsilon/2)^2 > 0 = f(0,0) \text{ for any } \epsilon > 0$$

$$f(\epsilon/2, 0) = 0^2 - (\epsilon/2)^2 = -(\epsilon/2)^2 < 0 = f(0,0) \text{ for any } \epsilon > 0$$

Therefore $(0,0)$ is a Saddle point of f .

Suppose that the second order partial derivatives of a function f are continuous in a disk at (x_0, y_0) and that $f_x(x_0, y_0) = 0 = f_y(x_0, y_0)$.

Let $\therefore$

$$A = \frac{\partial^2 f}{\partial x^2}(x_0, y_0) = f_{xx}(x_0, y_0)$$

$$B = \frac{\partial^2 f}{\partial x \partial y}(x_0, y_0) = f_{xy}(x_0, y_0)$$

$$C = \frac{\partial^2 f}{\partial y^2}(x_0, y_0) = f_{yy}(x_0, y_0)$$

$$\Delta = AC - B^2$$

$\begin{bmatrix} A & B \\ B & C \end{bmatrix}$ is called Hessian of f at (x_0, y_0)

(a) If $\Delta > 0$ and $A > 0$, then f has a local minimum at (x_0, y_0)

(b) If $\Delta > 0$ and $A < 0$, then f has a local maximum at (x_0, y_0)

(c) If $\Delta < 0$, then f has a Saddle point at (x_0, y_0)

Note:- If $\Delta = 0$, no conclusion can be drawn.
i.e., the test provides no information.

Ex 02:-

$$\text{Let } f(x, y) = x^3 + 3xy^2 - 3x^2 - 3y^2 + 4$$

Find and classify the critical points of f .

The critical points of f are given by

$$\frac{\partial f}{\partial x} = 0 \quad \text{and} \quad \frac{\partial f}{\partial y} = 0$$

$$\frac{\partial f}{\partial x} = 3x^2 + 3y^2 - 6x = 0 \quad \text{and} \quad \frac{\partial f}{\partial y} = 6xy - 6y = 0.$$

$$\Rightarrow x^2 + y^2 - 2x = 0 \quad \text{and} \quad y(x-1) = 0$$

* Do not divide by y .
 $y=0, x-1 \neq 0$
 $x-1=0, y \neq 0$
 $y=0$ and $x-1=0$

Case 01 :- $x^2 + y^2 - 2x = 0$ and $y = 0$.

$$\Rightarrow x^2 - 2x = 0$$

$$x(x-2) = 0$$

$$x=0 \quad \text{or} \quad x=2.$$

Then the Critical points of f are $(0,0), (2,0)$

Case 02 :-

$$x^2 + y^2 - 2x = 0 \quad \text{and} \quad x=1$$

$$1^2 + y^2 - 2(1) = 0$$

$$y^2 - 1 = 0$$

$$(y-1)(y+1) = 0$$

$$y=1 \quad \text{or} \quad y=-1$$

Therefore the Critical points of f are $(1,1), (1,-1)$

Altogether, there are 4 Critical points.

At a point (x,y) ,

$$A = \frac{\partial^2 f}{\partial x^2} = \frac{\partial}{\partial x} (3x^2 + 3y^2 - 6x)$$

$$= 6x - 6 = 6(x-1)$$

$$B = \frac{\partial^2 f}{\partial x \partial y} = \frac{\partial}{\partial x} (6xy - 6y) = 6y$$

$$C = \frac{\partial^2 f}{\partial y^2} = \frac{\partial}{\partial y} (6xy - 6y) = 6(x-1)$$

$$\Delta = AC - B^2$$

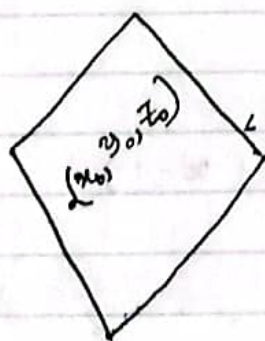
Critical Points	A	B	C	$\Delta = AC - B^2$	nature of the Critical Point
(0, 0)	-6	0	-6	$36 > 0$	Local maximum
(2, 0)	6	0	6	$36 > 0$	Local minimum.
(1, 1)	0	6	0	$-36 < 0$	Saddle point.
(1, -1)	0	-6	0	$-36 < 0$	Saddle point.

Homework:- Let $f(x, y) = 8x^4, 12x^2y, -9x^3$ find and Classify the Critical point of f .

- Lagrange's Multipliers -

This is a method finding maxima or minima of a function of several variables Subjective to a Constraint (a side Condition).

Suppose that we have to find the extrema (Maxima or Minima) of $f = f(x, y, z)$ Subject to the Condition (Constraint) $g(x, y, z) = 0$.



$g(x, y, z) = 0$
(Surface S)

Suppose that f has an extrema at (x_0, y_0, z_0) on S

Let $F = f + \lambda g$

Then we may have such that $\nabla F(x_0, y_0, z_0) = 0$

$$\nabla f(x, y, z) = \text{grad } F(x, y, z) .$$

$$= \frac{\partial F}{\partial x} \underline{i} + \frac{\partial F}{\partial y} \underline{j} + \frac{\partial F}{\partial z} \underline{k} .$$

Now ① $\Rightarrow$

$$\frac{\partial}{\partial x} (F + \lambda g) \underline{i} + \frac{\partial}{\partial y} (F + \lambda g) \underline{j} + \frac{\partial}{\partial z} (F + \lambda g) \underline{k} = 0$$

Since $\underline{i}$, $\underline{j}$, and $\underline{k}$ are linearly independent, we have

$$f_x + \lambda g_x = 0$$

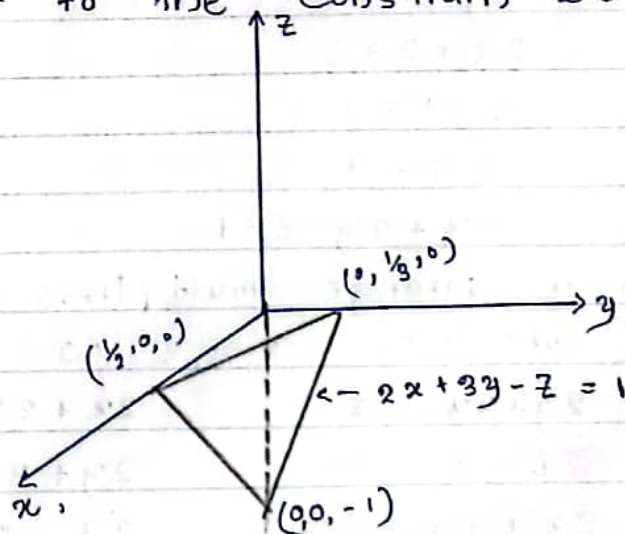
$$f_y + \lambda g_y = 0$$

$$f_z + \lambda g_z = 0 .$$

at (x_0, y_0, z_0) and $g(x_0, y_0, z_0) = 0$.

Ex 03:- Find the minimum value of $f(x, y, z) = x^2 + y^2 + z^2$ subject to the constraint $2x + 3y - z = 1$

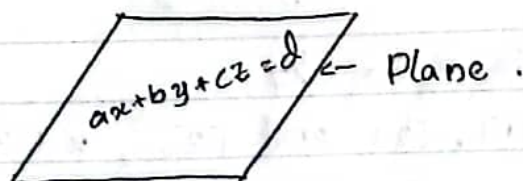
x	y	z
0	0	-1
1/2	0	0
0	1/3	0



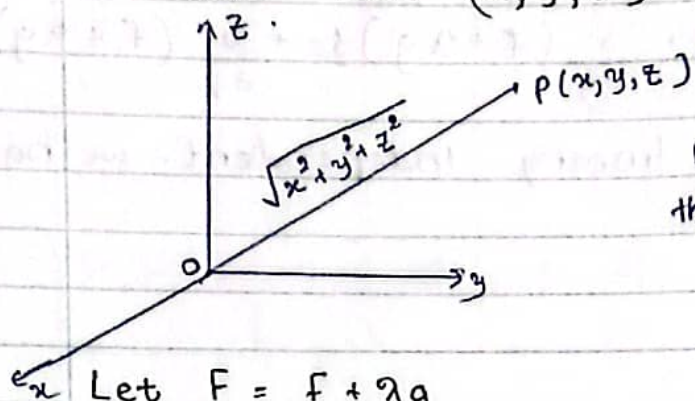
Note:- $ax + by + cz = d$

represents a plane in xyz -space.

Here a, b, c and d are constraints.



Let $g(x, y, z) = 2x + 3y - z - 1$
 $f(x, y, z) = x^2 + y^2 + z^2 = (\text{distance})^2$
 from the origin $(0, 0, 0)$ is to be minimized
 assumin that (x, y, z) lies on $g(x, y, z) = 0$



Maximum, doesn't exist b.
 the minimum exists.

Let $F = f + \lambda g$
 f is to be minimized or maximized
 g is constrain.

where $\lambda \neq 0$.

$$F = x^2 + y^2 + z^2 + \lambda(2x + 3y - z - 1)$$

$$\Rightarrow F_x = 2x + 2\lambda$$

$$F_y = 2y + 3\lambda$$

$$F_z = 2z - \lambda$$

$$F_\lambda = 2x + 3y - z - 1$$

Now by lagrange multiplies at the minimum of f (Subject to $g(x, y, z) = 0$) we have

~~$$2x + 2\lambda = 0$$~~

~~$$2z - \lambda = 0$$~~

~~$$2x + 3y - z = 1$$~~

$$2x + 2\lambda = 0 \Rightarrow x = -\lambda \quad \text{--- (1)}$$

$$2y + 3\lambda = 0 \Rightarrow y = -3\lambda/2 \quad \text{--- (2)}$$

$$2z - \lambda = 0 \Rightarrow z = \lambda/2 \quad \text{--- (3)}$$

$$2x + 3y - z = 1 \quad \text{--- (4)}$$

Substitute x, y , and z into [4] to obtain λ
 so we have, $2(-\lambda) + 3(-3\lambda/2) - \lambda/2 = 1$

$$\text{So } \lambda = -1/7$$

So ~~x~~ Then from (1), (2) and (3), $x + 3y - z = 1$
 we have $x = 1/7$, $y = 3/14$, $z = -1/14$

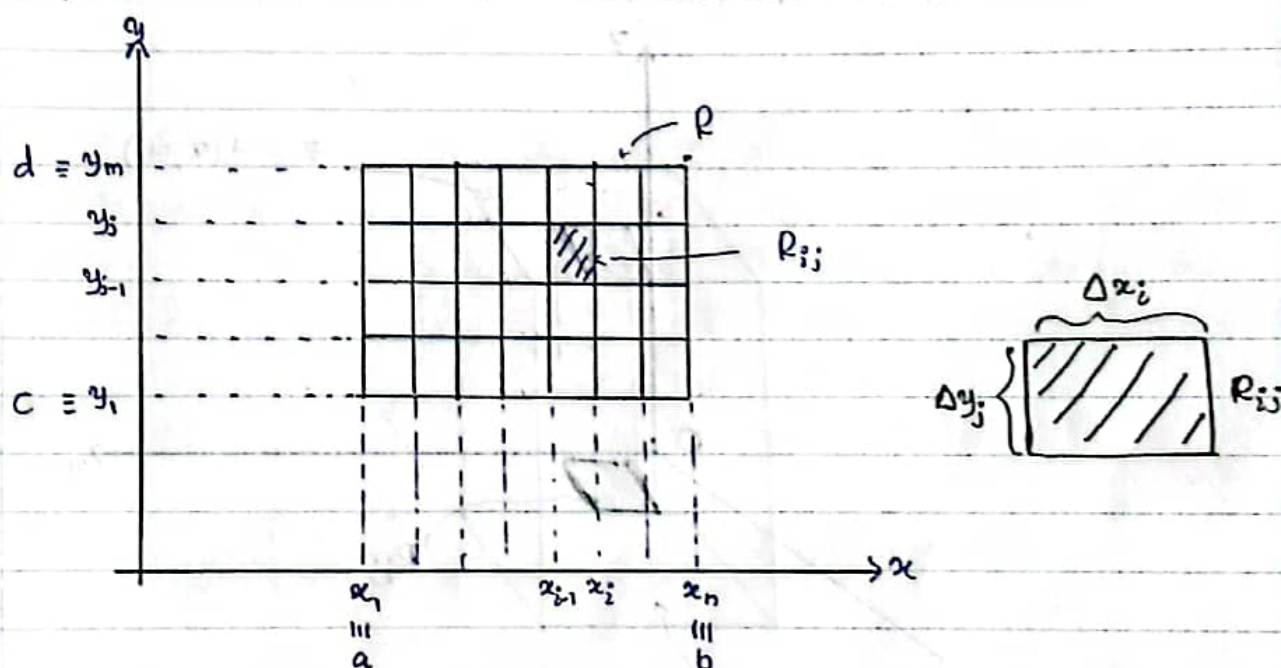
Min of f occurs at this point.

CHAPTER 06 - MULTIPLE INTEGRALS

$$\text{Let } R = \{ (x, y) \in \mathbb{R}^2 \mid a \leq x \leq b, c \leq y \leq d \}$$

Let $f: R \rightarrow \mathbb{R}$ be a bounded function. Consider the partition P of R into $m \times n$ rectangles created by the partitions

$\{x_1, x_2, \dots, x_n\}$ of $[a, b]$ and $\{y_1, y_2, \dots, y_m\}$ of $[c, d]$

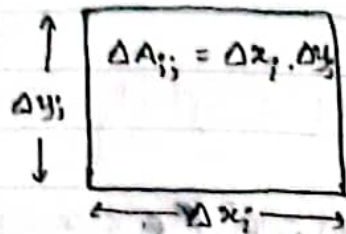


$$\text{Let } R_{ij} = \{ (x, y) \in \mathbb{R}^2 \mid x_{i-1} \leq x \leq x_i, y_{j-1} \leq y \leq y_j \}$$

for $i = 1, 2, 3, \dots, n$ and
 $j = 1, 2, 3, \dots, m$.

$$\text{Let } \Delta x_i = x_i - x_{i-1} \quad \text{for } i = 1, 2, \dots, n \text{ and}$$

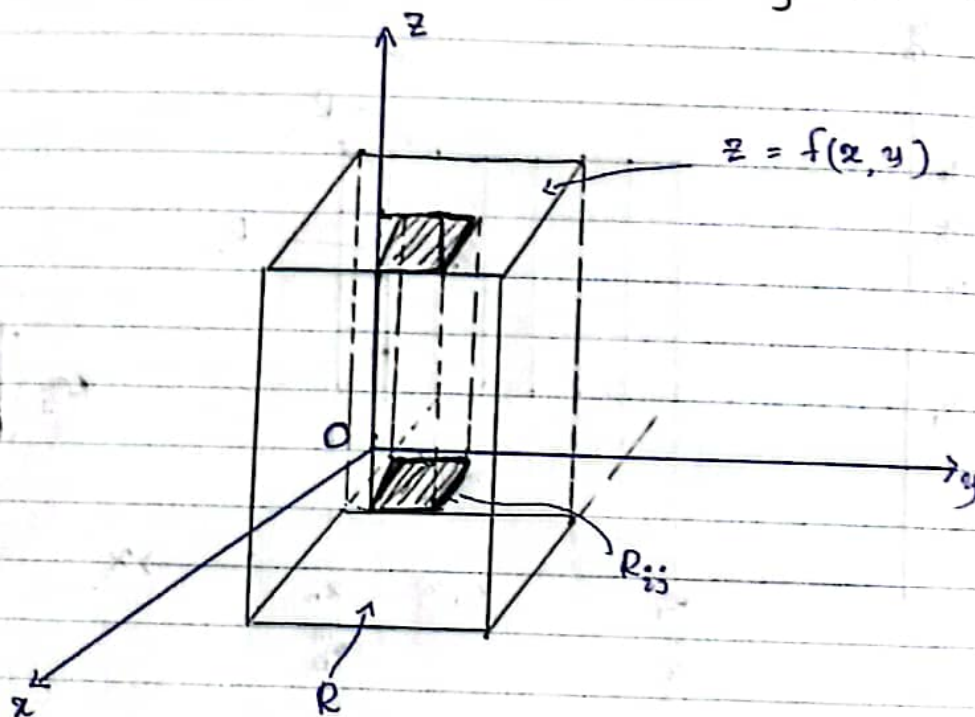
$$\Delta y_j = y_j - y_{j-1} \quad \text{for } j = 1, 2, \dots, m$$



The area of $R_{ij} = \Delta A_{ij} = \Delta x_j \times \Delta y_i$
for $i = 1, 2, \dots, n$ and $j = 1, 2, \dots, m$

$$\text{Let } m_{ij} = \inf \{ f(x, y) / (x, y) \in R_{ij} \}$$

$$M_{ij} = \sup \{ f(x, y) / (x, y) \in R_{ij} \}$$



Now from the upper Sum

$$U(f, P) = \sum_{i=1}^n \sum_{j=1}^m M_{ij} \Delta A_{ij} \text{ and}$$

The Lower Sum

$$L(f, P) = \sum_{i=1}^n \sum_{j=1}^m m_{ij} \Delta A_{ij}$$

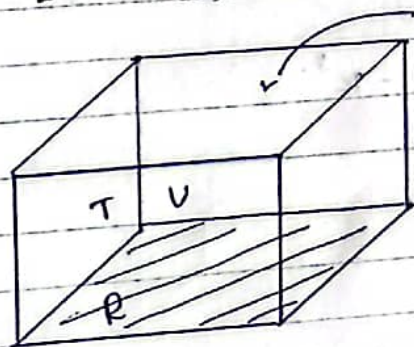
Let $u(f) = \inf \{ u(f, P) / P \text{ is a partition of } R \}$
 and $L(f) = \sup \{ L(f, P) / P \text{ is a partition of } R \}$

If $u(f) = L(f)$, then f is said to be Riemann integrable over R .

This Common Value is denoted by $\iint_R f(x, y) dA$ or $\iint_R f(x, y) dx dy$

Note :-

If $f(x, y) \geq 0$ for $(x, y) \in R$ and f is Riemann integrable over R , then the Volume of the Solid that lies above R and under the Surface $z = f(x, y)$ is defined by $\iint_R f(x, y) dA$.



$$V = \iint_R f(x, y) dA.$$

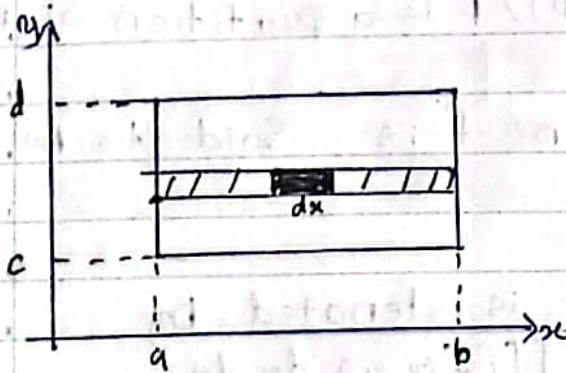
Theorem 01 :-

Let f be a Riemann integrable over the rectangle $R = \{ (x, y) \in \mathbb{R}^2 / a \leq x \leq b, c \leq y \leq d \}$
 $= [a, b] \times [c, d]$

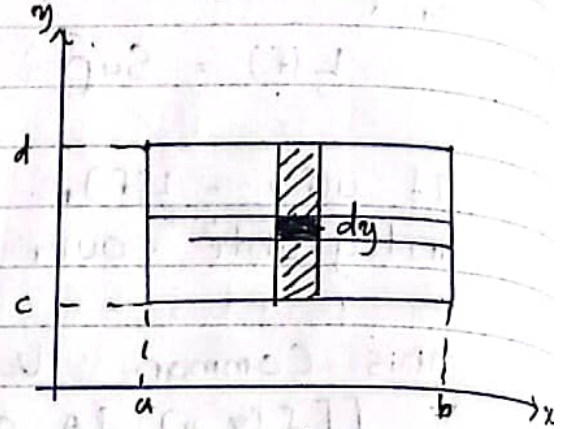
$$\text{Then, } \iint_R f(x, y) dA = \int_c^d \left[\int_a^b f(x, y) dx \right] dy$$

Min of f occurs at this point.

$$\iint_R f(x, y) dA = \int_a^b \left[\int_c^d f(x, y) dy \right] dx$$



$$\int_c^d \left[\int_a^b f(x, y) dx \right] dy$$



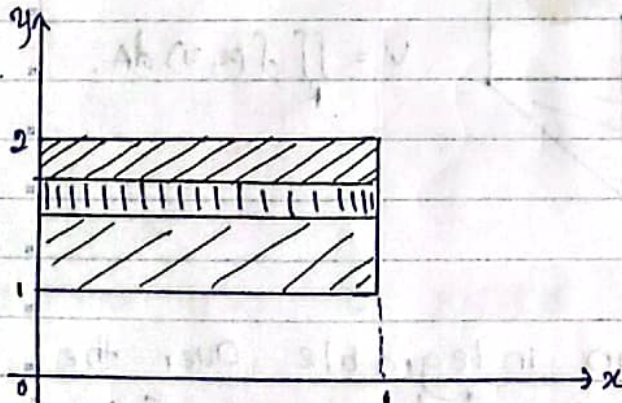
$$\int_a^b \left[\int_c^d f(x, y) dy \right] dx$$

Example :-

Evaluate $\iint_R (xy - x^3) dA$, where

$$R = \{(x, y) \in \mathbb{R}^2 / 0 \leq x \leq 1, 1 \leq y \leq 2\}$$

* Let $f(x, y) = xy - x^3$



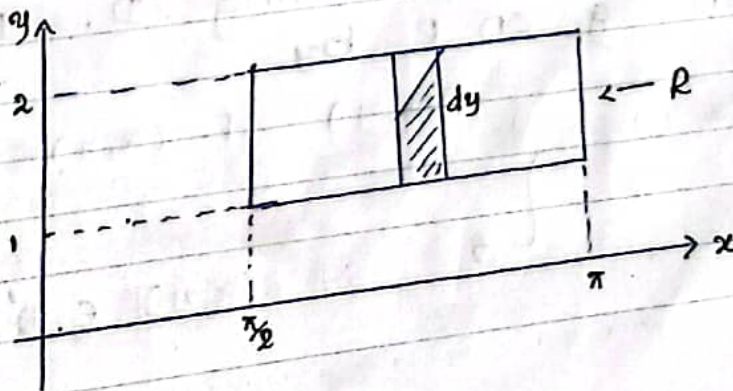
By theorem 1, we have,

$$\begin{aligned}
 \iint_R f(x, y) dA &= \int_{y=1}^2 \left[\int_{x=0}^1 f(x, y) dx \right] dy \\
 &= \int_{y=1}^2 \left[\int_{x=0}^1 (xy - x^3) dx \right] dy \\
 &= \int_{y=1}^2 \left[y \frac{x^2}{2} - \frac{x^4}{4} \right]_{x=0}^1 dy \\
 &= \int_{y=1}^2 \left(\frac{y}{2} - \frac{1}{4} - 0 \right) dy \\
 &= \left[\frac{y^2}{4} - \frac{y}{4} \right]_1^2 \\
 &= \frac{1}{4} \left[(2^2 - 2) - (1^2 - 1) \right] \\
 &= \frac{1}{2}
 \end{aligned}$$

Example :-

Evaluate $\iint_R x \sin(xy) dx dy$, where

$$R = \{(x, y) \in \mathbb{R}^2 / \frac{\pi}{2} \leq x \leq \pi, 1 \leq y \leq 2\}$$



... of f occurs of this point.

$$\begin{aligned}
 \iint_R x \sin(xy) \, dx \, dy &= \int_{x=\pi/2}^{\pi} \left[\int_{y=1}^2 x \sin(xy) \, dy \right] dx \\
 &= \int_{x=\pi/2}^{\pi} \left[-\cos(xy) \right]_{y=1}^2 dx \\
 &= \int_{x=\pi/2}^{\pi} (\cos x - \cos 2x) \, dx \\
 &= \left[\sin x - \frac{\sin 2x}{2} \right]_{x=\pi/2}^{\pi} \\
 &= \left(\sin \pi - \frac{\sin 2\pi}{2} \right) - \left(\sin \frac{\pi}{2} - \frac{\sin \pi}{2} \right) \\
 &= -1
 \end{aligned}$$

Double integration over general region.



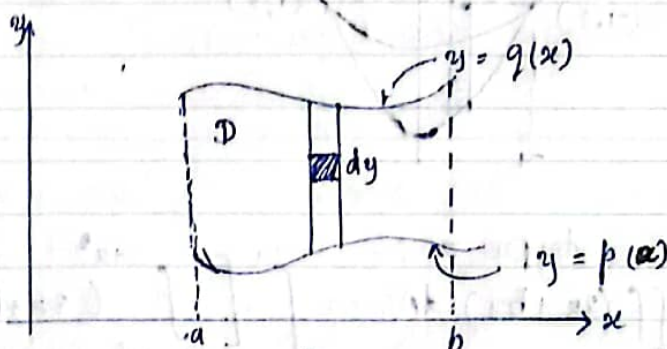
Let D be a bounded region. Then there exists a rectangle R containing D . Define a new function g on R by

$$g(x, y) = \begin{cases} f(x, y) & \text{if } (x, y) \in D \\ 0 & \text{if } (x, y) \in D' \end{cases}$$

If g is integrable over $\mathbb{R}^2$, then we say that f is integrable over D and we define the double integral of f over D by

$$\begin{aligned}\iint_D f(x, y) dA &= \iint_{\mathbb{R}^2} g(x, y) dA \\ &= \iint_{\mathbb{R}^2} g(x, y) dx dy.\end{aligned}$$

Type 01 :-



$$D = \left\{ (x, y) \in \mathbb{R}^2 \mid p(x) \leq y \leq q(x), a \leq x \leq b \right\}.$$

we may prove that

$$\iint_D f(x, y) dA = \int_a^b \left[\int_{y=p(x)}^{y=q(x)} f(x, y) dy \right] dx.$$

Example :-

Let D be the region bounded by the parabolas $y = 2x^2$ and $y = 1+x^2$.

Evaluate $\iint_D (3x + 6y) dA$.

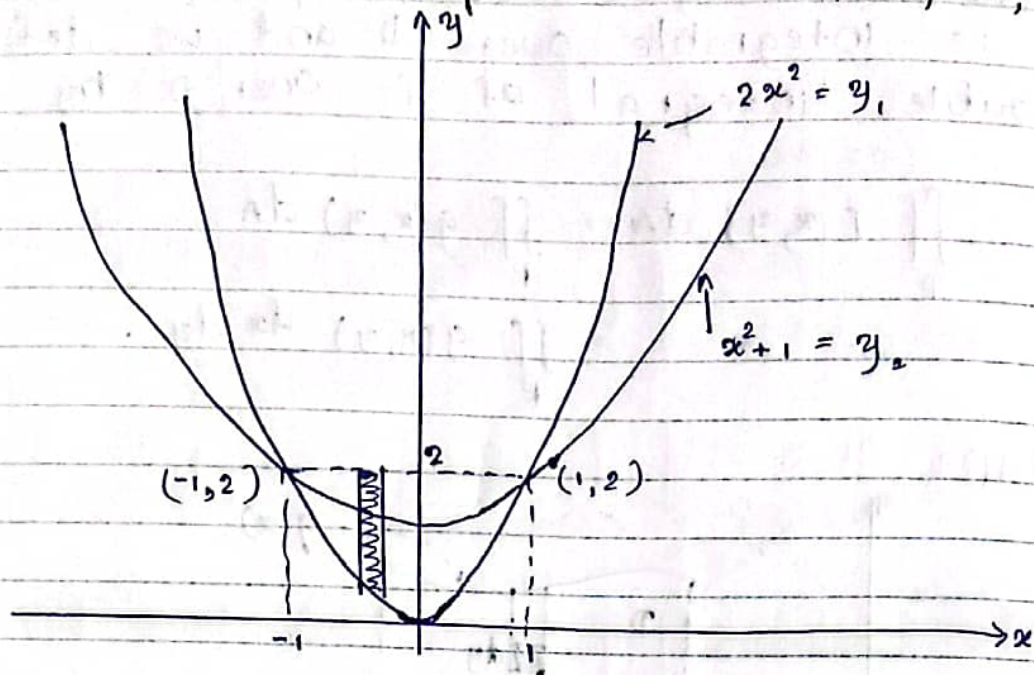
Let $y_1 = p(x) = 2x^2$ and $y_2 = q(x) = 1+x^2$

The points of the intersection of y_1 and y_2 are given by

$$y_1 = y_2 \Leftrightarrow 2x^2 = 1+x^2 \Leftrightarrow x = \pm 1$$

if $z = -1/19$
Min of f occurs at this point.

The intersection points are: $(1, 2)$, $(-1, 2)$



This is type of :-

$$\text{So, } \iint_D (3x + 5y) dA = \int_{x=-1}^1 \left[\int_{y=2x^2}^{1+x^2} (3x + 5y) dy \right] dx.$$

$$= \int_{x=-1}^1 \left[3xy + \frac{5}{2}y^2 \right]_{y=2x^2}^{1+x^2} dx.$$

$$= \int_{x=-1}^1 3 \left[\left\{ x(1+x^2) + \frac{5}{2}(1+x^2)^2 \right\} - \left\{ x \cdot 2x^2 + (2x^2)^2 \right\} \right] dx.$$

$$= \int_{x=-1}^1 3 (-3x^4 - x^3 + 2x^2 + x + 1) dx.$$

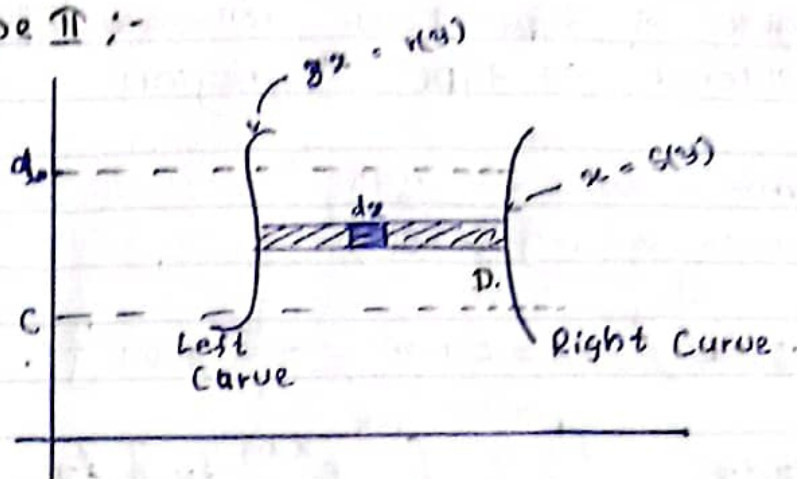
$$= 3 \left[-\frac{3x^5}{5} - \frac{x^4}{4} + \frac{2x^3}{3} + \frac{x^2}{2} + x \right]_{x=-1}^1$$

$$= 3 \left[\left(-\frac{3}{5} - \frac{1}{4} + \frac{2}{3} + \frac{1}{2} + 1 \right) - \left(\frac{3}{5} - \frac{1}{4} - \frac{2}{3} + \frac{1}{2} - 1 \right) \right]$$

$$= 3 \left[-\frac{6}{5} + \frac{4}{3} + 2 \right] = 3 \cdot \left(\frac{32}{15} \right)$$

$$= \frac{32}{5}$$

Type II :-



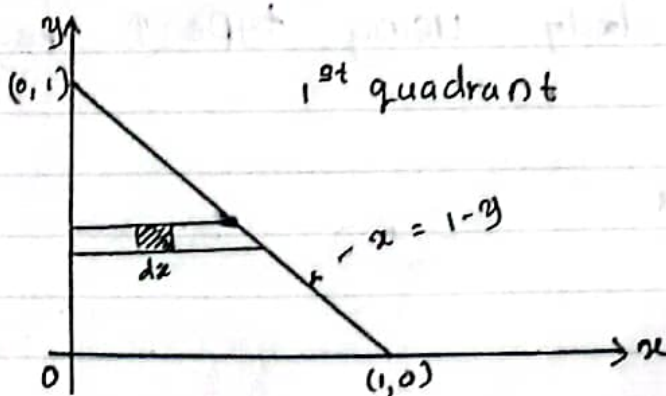
$r(y) \leq s(y) \quad \forall y \in [c, d]$ and r and s are Continuous on $[c, d]$

$D = \{ (x, y) \in \mathbb{R}^2 / r(y) \leq x \leq s(y), c \leq y \leq d \}$.
As in type I Case, we have: if f is Continuous on D , then

$$\iint_D f(x, y) dA = \int_{y=c}^d \left[\int_{x=r(y)}^{x=s(y)} f(x, y) dx \right] dy.$$

Example :-

Evaluate $\iint_D e^{x+y} dx dy$, where D is the region in the first quadrant bounded by the Co-ordinate axes and the line $x+y=1$



$$0 \leq x \leq 1-y.$$

This region is type I as well as type II and can be considered as type II region.

Left Curve: $x = 0 \leftarrow r(y)$

Right Curve: $x = 1 - y \leftarrow s(y)$

$$D = \{ (x, y) \in \mathbb{R}^2 / 0 \leq x \leq 1 - y, 0 \leq y \leq 1 \}.$$

$$\iint_D e^{x+y} dx dy = \int_{y=0}^1 \left[\int_{x=0}^{1-y} e^{x+y} dx \right] dy.$$

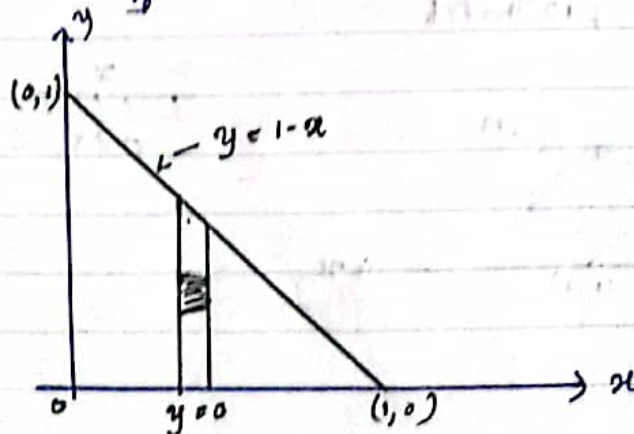
$$= \int_{y=0}^1 \left[\{ e^{x+y} \}_{x=0}^{1-y} \right] dy.$$

$$= \int_{y=0}^1 (e^{1-y+y} - e^{0+y}) dy.$$

$$= \int_{y=0}^1 (e - e^y) dy$$

$$= [ey - e^y]_{y=0}^1$$

Evaluate $\iint_D e^{x+y} dx dy$ using type II region



$$\iint_D e^{x+y} dy dx = \int_{x=0}^1 \left[\int_{y=0}^{1-x} e^{x+y} dy \right] dx.$$

$$= \int_0^1 \left[e^{x+y} \right]_{y=0}^{1-x} dx.$$

$$= \int_0^1 [e^{x+1-x} - e^{x+0}] dx.$$

$$= \int_0^1 (e - e^x) dx$$

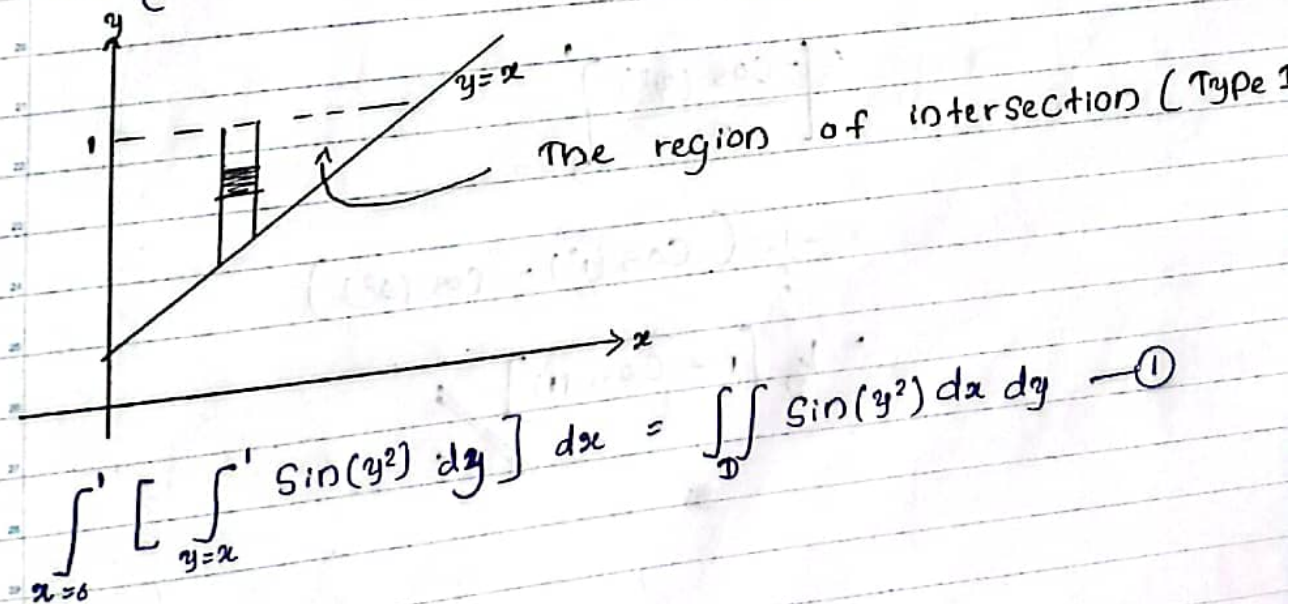
$$= (xe - e^x)_0^1.$$

$$= (e - e) - (0 - 1)$$

$$= 1$$

Example :- Evaluate $\int_0^1 \left[\int_x^1 \sin(y^2) dy \right] dx.$

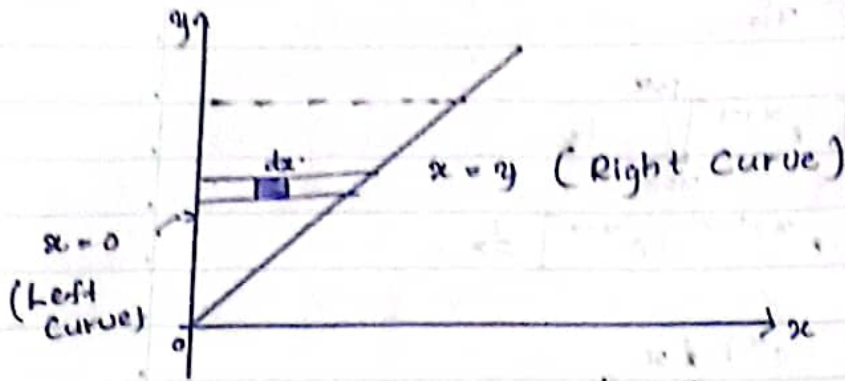
$$D = \{ (x,y) \in \mathbb{R}^2 / x \leq y \leq 1, 0 \leq x \leq 1 \}.$$



min of f occurs at this point.

1/4

Next Consider D as type II region.



$$\iint_D \sin(y^2) dx dy = \int_{y=0}^1 \left[\int_{x=0}^y \sin(y^2) dx \right] dy$$

$$= \int_{y=0}^1 \left[x \sin(y^2) \right]_0^y dy$$

$$= \int_{y=0}^1 (y \sin(y^2) - 0) dy$$

$$= \int_{y=0}^1 y \sin(y^2) dy$$

$$= \left[\frac{-\cos(y^2)}{2} \right]_{y=0}^1$$

$$= -\frac{1}{2} (\cos(1^2) - \cos(0^2))$$

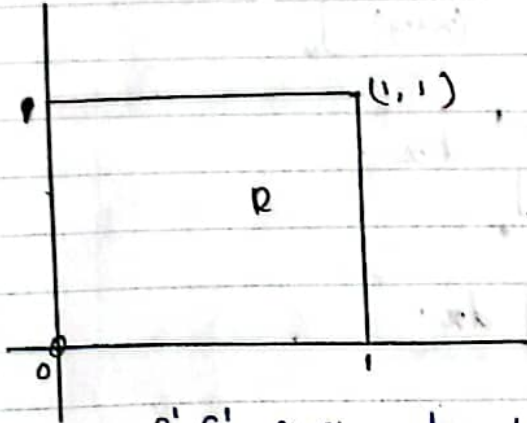
$$= \frac{1}{2} [1 - \cos(1)] //$$

Example :-

Show that $\int_0^1 \int_0^1 \frac{x-y}{(x+y)^3} dx dy = -\frac{1}{2}$ and

$$\int_0^1 \int_0^1 \frac{x-y}{(x+y)^3} dy dx = \frac{1}{2}.$$

$$D = \{(x, y) \in \mathbb{R}^2 / 0 \leq x \leq 1, 0 \leq y \leq 1\}, \{(0, 0)\}.$$



$$I_1 = \int_0^1 \int_0^1 \frac{x-y}{(x+y)^3} dx dy.$$

$$= \int_0^1 \int_0^1 \left(\frac{1}{(x+y)^2} - \frac{2y}{(x+y)^3} \right) dx dy.$$

$$= \int_0^1 \left[\int_0^1 \left(\frac{1}{(x+y)^2} - \frac{2y}{(x+y)^3} \right) dx \right] dy.$$

$$= \int_0^1 \left[\frac{-1}{x+y} - \frac{2y}{-2(x+y)^2} \right]_0^1 dy.$$

$$= \int_0^1 \left[\frac{-1}{1+y} + \frac{2}{(1+y)^2} - 0 \right] dy$$

$$= \int_0^1 \frac{-1}{(1+y)^2} dy$$

$$= - \int_0^1 (1+y)^{-2} dy$$

$$= \left[\frac{(1+y)^{-1}}{-1} \right]_0^1$$

$$= -\frac{1}{2}$$

$$I_2 = \int_0^1 \int_0^1 \left[\frac{2x}{(x+y)^3} - \frac{1}{(x+y)^2} \right] dy dx$$

$$= \int_0^1 \left[\int_0^1 \frac{2x}{(x+y)^3} - \frac{1}{(x+y)^2} dy \right] dx$$

$$= \int_0^1 \left[2x \frac{1}{-2(x+y)^2} + \frac{1}{(x+y)} \right]_0^1 dx$$

$$= \int_0^1 \left[\frac{-x}{(x+y)^2} + \frac{1}{(x+y)} \right]_0^1 dx$$

$$= \int_0^1 \left[\frac{1}{x+1} + \frac{x}{(x+1)^2} \right] dx$$

$$= \int_0^1 \frac{1}{(x+1)^2} dx$$

$$= \left[\frac{(x+1)^{-1}}{-1} \right]_0^1$$

$$= -\left(\frac{1}{2} - 1\right)$$

$$= \underline{\underline{\frac{1}{2}}}$$

Note that :-

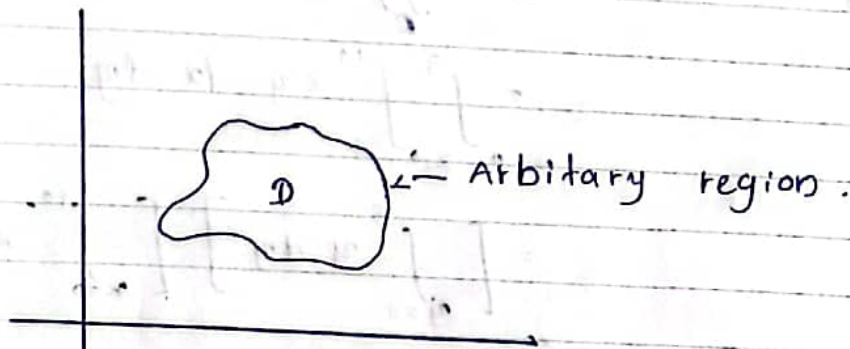
According to theorem!

$f(x, y) = \frac{x-y}{(x+y)^2}$ Can not be integrated over

$$R = \{(x, y) / 0 \leq x \leq 1, 0 \leq y \leq 1\}.$$

Note :-

(1)



The area of the region $D = \iint_D 1 \cdot dA$.

(2) If $f(x, y) = g(x)g(y)$, then

$$\int_a^b \int_c^d f(x, y) dy dx = \left[\int_a^b g(x) dx \right] \left[\int_c^d g(y) dy \right]$$

Example :-

A rectangular sheet lies in the region $0 \leq x \leq 10, 0 \leq y \leq 5$.

find the total mass of the sheet if the mass per unit area of the sheet is given by $m(x, y) = xy$.

$$m = m(x, y) = xy$$

$$dA = dx dy$$

∴ The mass of $dA = dA m$

$$= m dx dy$$

$$= xy dx dy$$

$$∴ \text{Total mass of } M = \iint_R m(x, y) dA$$

$$= \int_{y=0}^5 \int_{x=0}^{10} xy dx dy$$

$$= \left[\int_{y=0}^5 y dy \right] \left[\int_{x=0}^{10} x dx \right]$$

$$= \left[\frac{y^2}{2} \right]_0^5 \left[\frac{x^2}{2} \right]_0^{10}$$

$$= \frac{1}{2} [25] \cdot \frac{1}{2} [100]$$

$$= \underline{\underline{625 \text{ units}}}$$

Home work :-

- Q1) Find the volume of the Solid that lies under the paraboloid $z = x^2 + y^2$ and above the region D in the xy -Plane bounded by the lines $y = 2x$ and the Parabola $y = x^2$

[Answer: - $\frac{216}{35}$]

- Q2) Find the Volume of the Solid S that is bounded by the elliptic paraboloid $xy^2 + y^2x^2 + 2y^2 + z = 16$ the planes $x = 2$ and $y = 2$ and the Coordinate planes.

[Answer: - 48]

Triple Integrals :-

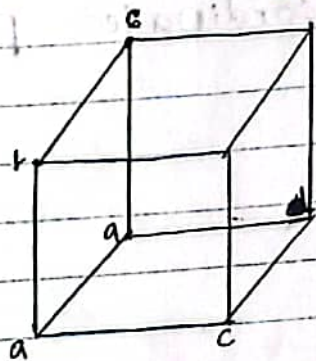
* Theorem 02 :-

If f is integrable over rectangular box B ,

$$B = \{ (x, y, z) \in \mathbb{R}^3 / a \leq x \leq b, c \leq y \leq d, r \leq z \leq s \}$$

Then

$$\iiint_B f(x, y, z) \, dV = \int_{z=r}^s \int_{y=c}^d \int_{x=a}^b f(x, y, z) \, dx \, dy \, dz.$$



Example :-

Evaluate $\iiint_B (x+y+z) \, dV$, where

$$B = \{ (x, y, z) / 0 \leq x \leq 1, 1 \leq y \leq 2, 0 \leq z \leq 1 \}.$$

$$\iiint_B (x+y+z) \, dV = \int_{z=0}^1 \int_{y=1}^2 \int_{x=0}^1 (x+y+z) \, dx \, dy \, dz$$

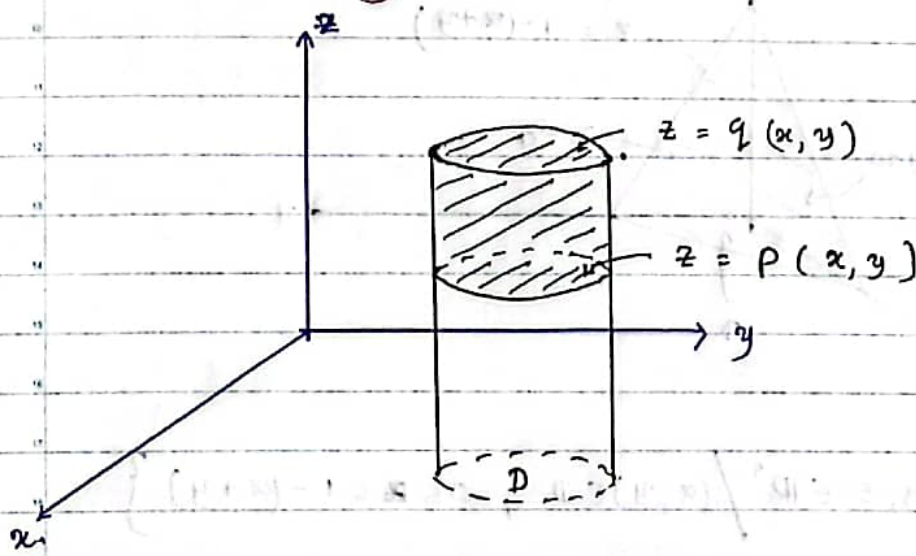
$$= \int_{z=0}^1 \int_{y=1}^2 \left[\frac{x^2}{2} + xyz \right]_{x=0}^1 dy \, dz.$$

$$= \int_{z=0}^1 \int_{y=1}^2 \left[\frac{1}{2} + yz \right] dy \, dz$$

$$= \int_{z=0}^1 \left[\frac{y}{2} + \frac{y^2 z}{2} \right]_{y=1}^2 dz.$$

$$\begin{aligned}
 &= \frac{1}{2} \left[z + \frac{3z^2}{2} \right]_{z=0}^1 \\
 &= \frac{1}{2} \left(1 + \frac{3}{2} \right) \\
 &= \underline{\underline{\frac{5}{4}}}
 \end{aligned}$$

General region:-



Here both $p(x, y)$ and $q(x, y)$ are Continuous in D , and $p(x, y) \leq q(x, y)$ $\forall (x, y) \in D$.

Let $w = \{(x, y, z) \in \mathbb{R}^3 \mid (x, y) \in D, p(x, y) \leq z \leq q(x, y)\}$

Let $f = f(x, y, z)$ be a function which is Continuous in w .

We shall assume that $p(x, y)$ and $q(x, y)$ are Continuous in D .

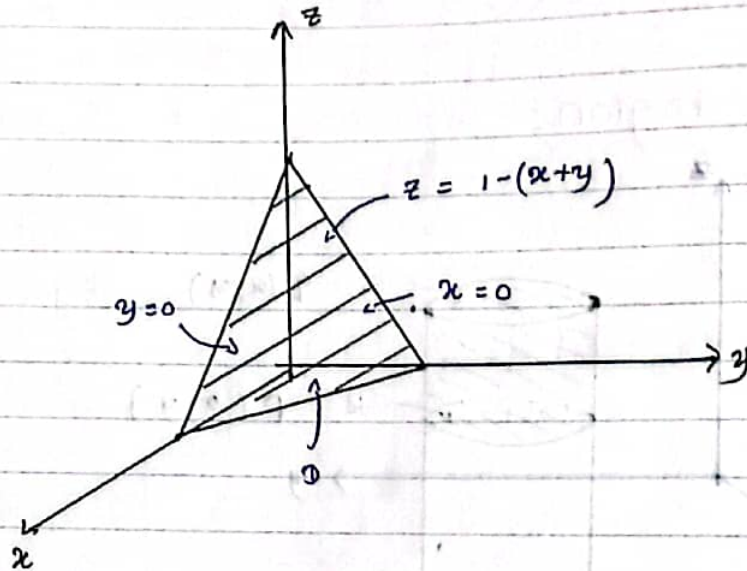
Then,

$$\iiint_w f(x, y, z) \, dv = \iint_D \left[\int_{z=p(x, y)}^{z=q(x, y)} f(x, y, z) \, dz \right] dA.$$

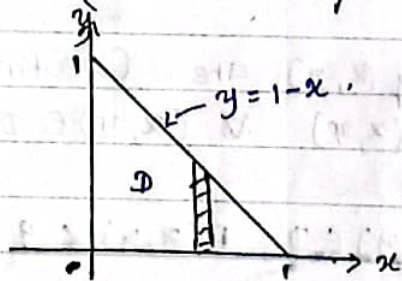
Example :-

Find $\iiint_w xz \, dv$, where w is a solid tetrahedron bounded by the planes :-

$$x=0, y=0 \text{ and } x+y+z=1$$



$$w = \{(x, y, z) \in \mathbb{R}^3 \mid (x, y) \in D, 0 \leq z \leq 1 - (x+y)\}$$



$$D = \{(x, y) \mid 0 \leq y \leq 1-x, 0 \leq x \leq 1\}$$

$$\iiint_w xz \, dv = \iint_D \left[\int_{z=0}^{1-(x+y)} xz \, dz \right] dA$$

$$= \iint_D \left[\frac{xz^2}{2} \right]_{z=0}^{1-(x+y)} dA$$

$$= \iint_D x \left[\frac{1-(x+y)^2}{2} \right] dA$$

$$= \int_{x=0}^1 \int_{y=0}^{1-x} \frac{x}{2} [1 - (x+y)^2] dy \, dx$$

$$= \int_{x=0}^1 \frac{x}{2} \left[\frac{(1-x-y)^3}{-3} \right]_{y=0}^{1-x} dx.$$

$$= \int_{x=0}^1 \frac{x}{6} [(1-x)^3] dx.$$

$$= \frac{1}{6} \int_0^1 [(1-x)^3 - (1-x)^4] dx$$

$$= \frac{1}{6} \left[\frac{(1-x)^4}{-4} - \frac{(1-x)^5}{(-5)} \right]_{x=0}^1$$

$$= \frac{1}{6} \left(\frac{1}{4} - \frac{1}{5} \right)$$

$$= \frac{1}{120}$$

Homework:-

Evaluate $\iiint_E z \, dv$, where E is solid tetrahedron bounded by the four planes:

$x=0$, $y=0$, $z=0$ and $x+y+z=1$.

change of variables in multiple integrals

- when $n=1$;

Let f be continuous on $[a, b]$. Let $x = g(y)$. If g and g' are continuous on $[c, d]$, then

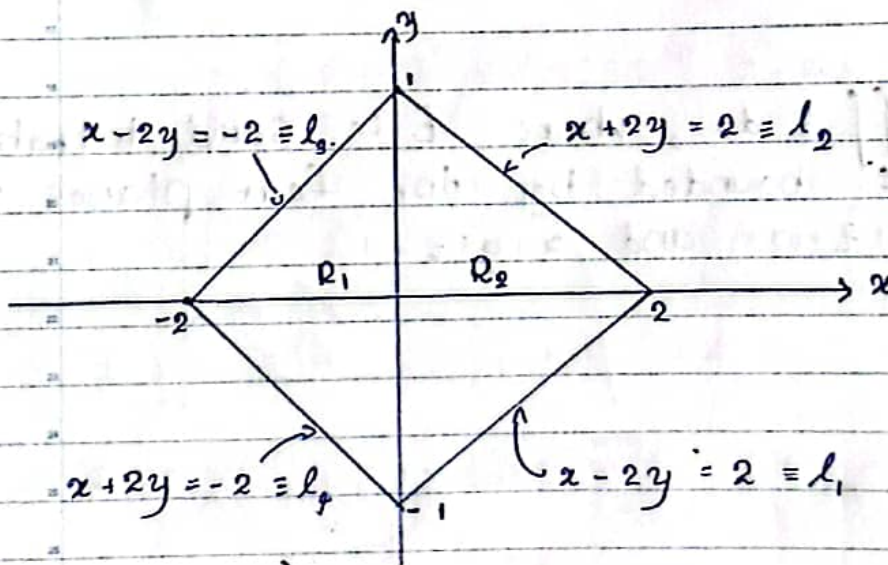
$$\int_a^b f(x) dx = \int_c^d f(g(y)) g'(y) dy$$

- when $n=2$;

Example:-

Evaluate $\iint_R (3x+by)^2 dA$, where R is the region bounded by the lines:

$$x+y=2, \quad x-2y=2, \quad x+2y=-2, \quad \text{and} \quad x-2y=-2$$



$$R = R_1 \cup R_2$$

$$\iint_R (3x+by)^2 dA = \iint_{R_1} (3x+by)^2 dA + \iint_{R_2} (3x+by)^2 dA$$

$$= \int_{x=-2}^0 \int_{y=-\frac{x+2}{2}}^{\frac{x+2}{2}} (3x+by)^2 dy dx + \int_{x=0}^2 \int_{y=\frac{x-2}{2}}^{\frac{x-2}{2}} (3x+by)^2 dy dx$$

These integrals can be evaluated but it is tedious. we ~~the~~ shall try Change of variables.

Let us try the Change of variables given by $u = x + 2y$ and $v = x - 2y$.

on l_1 ;

$$x - 2y = 2 \Rightarrow v = 2$$

on l_2 ;

$$x + 2y = 2 \Rightarrow u = 2$$

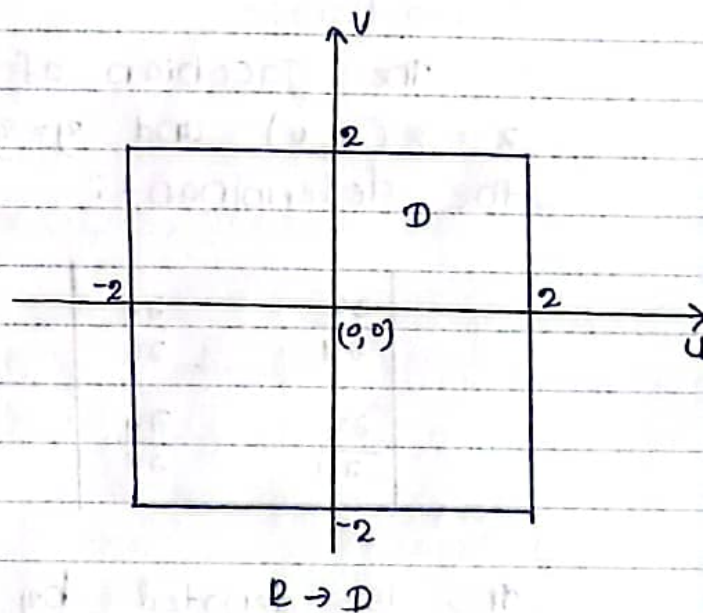
on l_3

$$x - 2y = -2 \Rightarrow v = -2$$

on l_4

$$x + 2y = -2 \Rightarrow u = -2$$

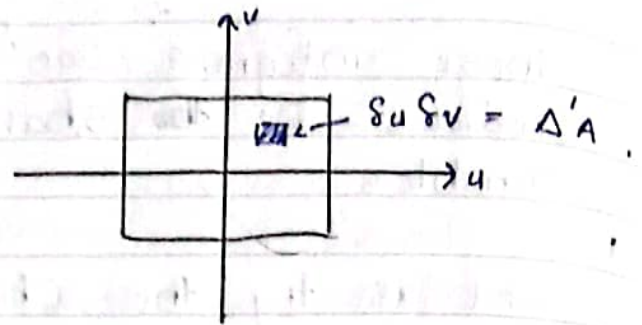
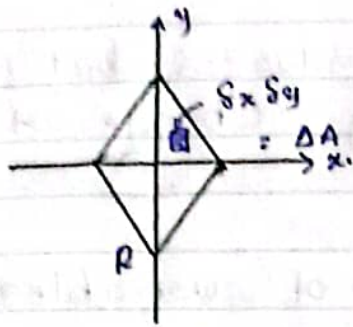
$$(0,0) \rightarrow (0,0)$$



$$\text{Also, } x = \frac{u+v}{2} \text{ and } y = \frac{u-v}{2}$$

under this transformation, the region R in the xy -plane gets mapped to the rectangular region D .

$$\begin{aligned} f(x,y) &= (3x+by)^2 \\ &= a(x+2y)^2 \\ &= au^2 \end{aligned}$$



$$\iint_R (3x+by)^2 dA = \iint_D q u^2 [\quad] - (*)$$

How are $\Delta A = \delta x \delta y$ and $\Delta A' = \delta u \delta v$ related?
Here we need the Jacobian of the transformation

Definition:-

The Jacobian of the transformation given $x = x(u, v)$ and $y = y(u, v)$ is defined to be the determinant:

$$\begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix}$$

This is denoted by $\frac{\partial(x, y)}{\partial(u, v)}$ or $J(x, y)$

$$\text{So, } \frac{\partial(x, y)}{\partial(u, v)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix}$$

$$= \frac{\partial x}{\partial u} \cdot \frac{\partial y}{\partial v} - \frac{\partial x}{\partial v} \cdot \frac{\partial y}{\partial u}$$

$$\text{Also, } \frac{\partial(x, y)}{\partial(u, v)} = \frac{1}{\frac{\partial(u, v)}{\partial(x, y)}}$$

Theorem;

Let $z = f(x, y)$ be a Continuous function on a region R in the xy -plane.

Let T be a one to one transformation that maps the region D in the uv -plane onto R under the change of variables $x = x(u, v)$ and $y = y(u, v)$ where x and y are continuously differentiable functions.

If $\frac{\partial(x, y)}{\partial(u, v)} \neq 0$, then

$$\iint_R f(x, y) dA = \iint_D f(x(u, v), y(u, v)) \left| \frac{\partial(x, y)}{\partial(u, v)} \right| du dv.$$

Let us get back to the last example.

$$u = x + 2y \Rightarrow \frac{\partial u}{\partial x} = 1 \text{ and } \frac{\partial u}{\partial y} = 2$$

$$v = x - 2y \Rightarrow \frac{\partial v}{\partial x} = 1 \text{ and } \frac{\partial v}{\partial y} = -2.$$

$$\therefore \frac{\partial(u, v)}{\partial(x, y)} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix}$$

$$= \begin{vmatrix} 1 & 2 \\ 1 & -2 \end{vmatrix} = -2 - 2 = -4.$$

$$\Rightarrow \frac{\partial(x, y)}{\partial(u, v)} = \frac{1}{\frac{\partial(u, v)}{\partial(x, y)}} = \frac{-1}{4} \neq 0.$$

$$\text{Now } \iint_R f(x, y) dA = \iint_D f(x(u, v), y(u, v)) \left| \frac{\partial(x, y)}{\partial(u, v)} \right| du dv$$

$$\Rightarrow \iint_R (3x + 4y)^2 dA = \iint_D 9u^2 \left| \frac{-1}{4} \right| du dv$$

$$= \int_{v=-2}^2 \left[\int_{u=-2}^2 \frac{9u^2}{4} du \right] dv$$

$$= \int_{v=-2}^2 \frac{9}{4} \left[\frac{u^3}{3} \right]_{u=-2}^2 dv$$

$$= \int_{v=-2}^2 \frac{9}{4} \left(\frac{8}{3} + \frac{8}{3} \right) dv$$

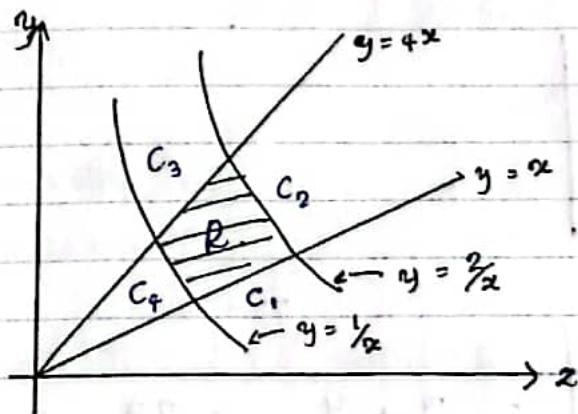
$$= \int_{v=-2}^2 12 dv$$

$$= 12(2+2)$$

$$= \underline{\underline{48}}$$

Example :-

Evaluate $\iint_R x^2 y^2 dA$ where R is the region in the 1st quadrant bounded by $xy=1$, $xy=2$, $y=x$, and $y=4x$.



on C_1 ;

$$y = x$$

$$\therefore u = x^2 \text{ and } v = \frac{x}{x} = 1$$

on C_2 ;

$$y = \frac{2}{x}$$

$$\therefore u = \frac{2}{x} \cdot x = 2 \text{ and } v = \frac{2/x}{x} = \frac{2}{x^2}$$

on C_3 ;

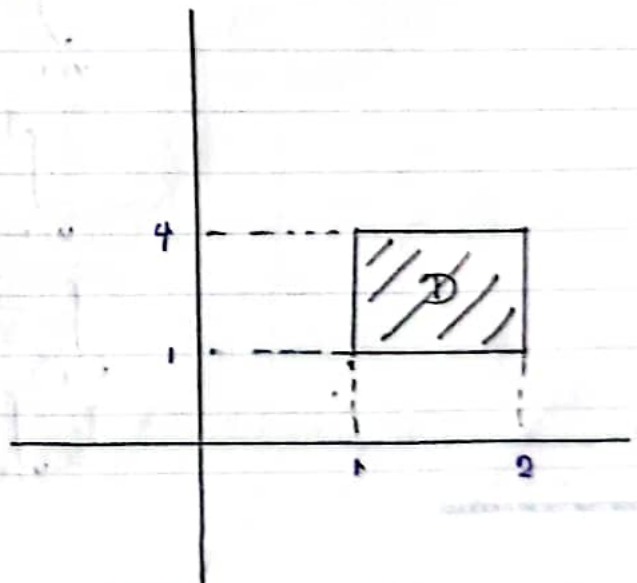
$$y = 4x$$

$$\therefore u = y/x = 4x/x = 4$$

on C_4 ;

$$y = 1/x$$

$$u = x \cdot \frac{1}{x} = 1$$



$$u = xy \Rightarrow \frac{\partial u}{\partial x} = y \text{ and } \frac{\partial u}{\partial y} = x.$$

$$v = \frac{y}{x} \Rightarrow \frac{\partial v}{\partial x} = -\frac{y}{x^2} \text{ and } \frac{\partial v}{\partial y} = \frac{1}{x}.$$

$$\frac{\partial(u, v)}{\partial(x, y)} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix}$$

$$= \begin{vmatrix} y & x \\ -\frac{y}{x^2} & \frac{1}{x} \end{vmatrix} = \frac{y}{x} + \frac{y}{x} = \frac{2y}{x}$$

$$= 2v$$

$$\therefore \left| \frac{\partial(x, y)}{\partial(u, v)} \right| = \frac{1}{\frac{\partial(u, v)}{\partial(x, y)}} = \frac{1}{2v} \neq 0.$$

$$\text{Then } \iint_R x^2 y^2 dA = \iint_D u^2 \left| \frac{1}{2v} \right| du dv$$

$$= \int_{v=1}^4 \left[\int_{u=1}^2 \frac{u^2}{2v} du \right] dv$$

$$= \int_{v=1}^4 \frac{1}{2v} \left[\frac{u^3}{3} \right] dv$$

$$= \int_{v=1}^4 \frac{1}{6v} (8-1) dv$$

$$= \frac{7}{6} \int_{v=1}^4 \frac{1}{v} dv.$$

$$= \frac{7}{6} [\ln |v|]_1^4$$

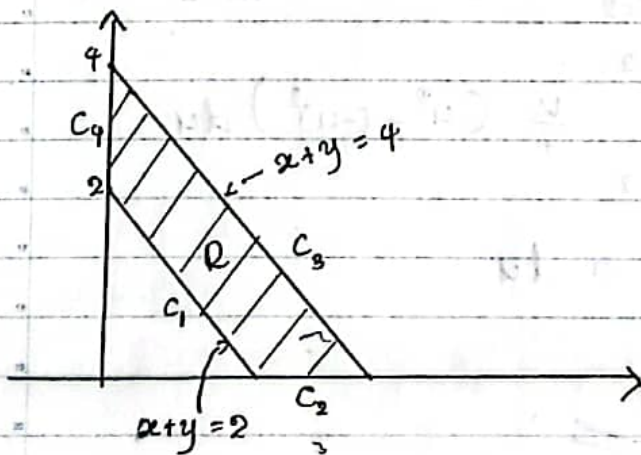
$$= \frac{7}{6} [\ln |4| - \ln |1|]$$

$$= \frac{7}{6} \ln |4|$$

Example:-

Evaluate $\iint_R \frac{y-x}{y+x} dA$, where R is the region

bounded by $x+y=4$, $x+y=2$, $x=0$ and $y=0$.



Let $u = x+y$ and $v = y-x$

on C_1 :- $x+y=2 \Rightarrow u=2$

on C_3 :- $x+y=4 \Rightarrow u=4$

on C_2 :- $y=0 \Rightarrow v=-u$

on C_4 :- $x=0 \Rightarrow v=u$

Also;

$$\frac{\partial u}{\partial x} = 1, \quad \frac{\partial u}{\partial y} = 1$$

$$\frac{\partial v}{\partial x} = -1, \quad \frac{\partial v}{\partial y} = 1$$

$$\frac{\partial (u,v)}{\partial (x,y)} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix}$$

$$= \begin{vmatrix} 1 & 1 \\ -1 & 1 \end{vmatrix} = 1+1=2$$

$$\therefore \frac{\partial(x, y)}{\partial(u, v)} = \frac{1}{\frac{\partial(u, v)}{\partial(x, y)}} = \frac{1}{2} \neq 0$$

$$\therefore \iint_R \frac{y-x}{y+x} dA = \iint_D \frac{v}{u} \left| \frac{1}{2} \right| du dv$$

$$= \int_{u=-2}^2 \left[\int_{v=-u}^u \frac{v}{2u} dv \right] du$$

$$= \int_{u=-2}^2 \frac{v}{2} \left[\ln \frac{v^2}{2} \right]_{v=-u}^u du$$

$$= \int_{u=-2}^2 \frac{v}{4} (u^2 - (-u)^2) du$$

$$= \int_{u=-2}^2 0 du$$

$$= 0$$

Homework:-

Evaluate $\iint_R [(x-y)^2 + 2] e^{x+y} dx dy$, where R is the region bounded by the lines $x+y=2$, $x=0$, and $y=0$.

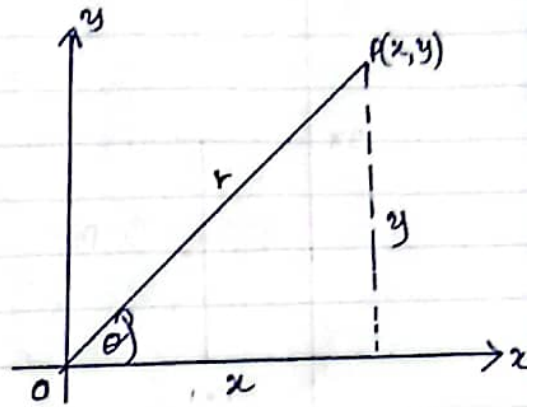
Polar Coordinates :-

* Plane Coordinates :-

$$OP = r > 0$$

$$x = r \cos \theta$$

$$y = r \sin \theta ; 0 \leq \theta \leq 2\pi$$

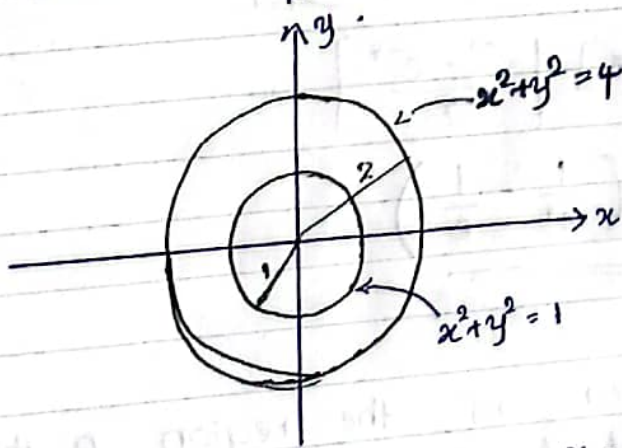


$$\frac{\partial(x, y)}{\partial(r, \theta)} = \begin{vmatrix} \frac{\partial x}{\partial r} & \frac{\partial y}{\partial r} \\ \frac{\partial x}{\partial \theta} & \frac{\partial y}{\partial \theta} \end{vmatrix} = \begin{vmatrix} \cos \theta & \sin \theta \\ -r \sin \theta & r \cos \theta \end{vmatrix}$$
$$= r \cos^2 \theta - (-r \sin^2 \theta)$$

$$= r$$

Example :-

Evaluate $\iint_R e^{-(x^2+y^2)} dA$, where $R = \{(x, y) \in \mathbb{R}^2 \mid 1 \leq x^2 + y^2 \leq 4\}$.



use polar coordinates, $x = r \cos \theta$ and $y = r \sin \theta$

$$\Rightarrow x^2 + y^2 = r^2$$

$$\text{on } x^2 + y^2 = 1 ;$$

$$r^2 = 1 \Rightarrow r = 1$$

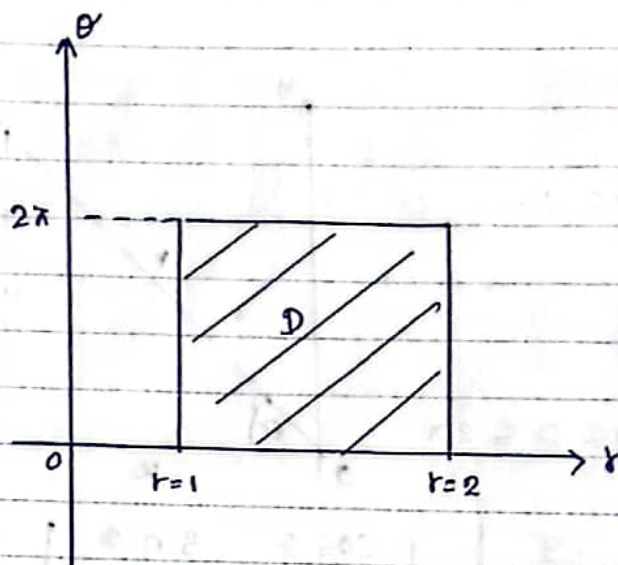
$$\theta : 0 \rightarrow 2\pi$$

$$\text{on } x^2 + y^2 = 4 ;$$

$$r^2 = 4 \Rightarrow r = 2$$

$$\theta : 0 \rightarrow 2\pi$$

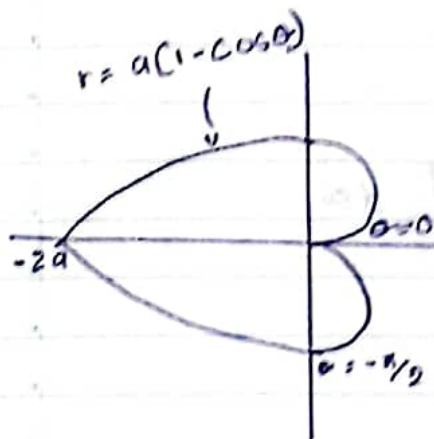
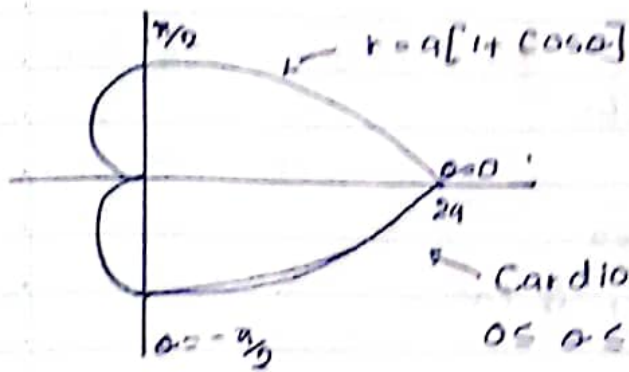
this occurs at
this point.



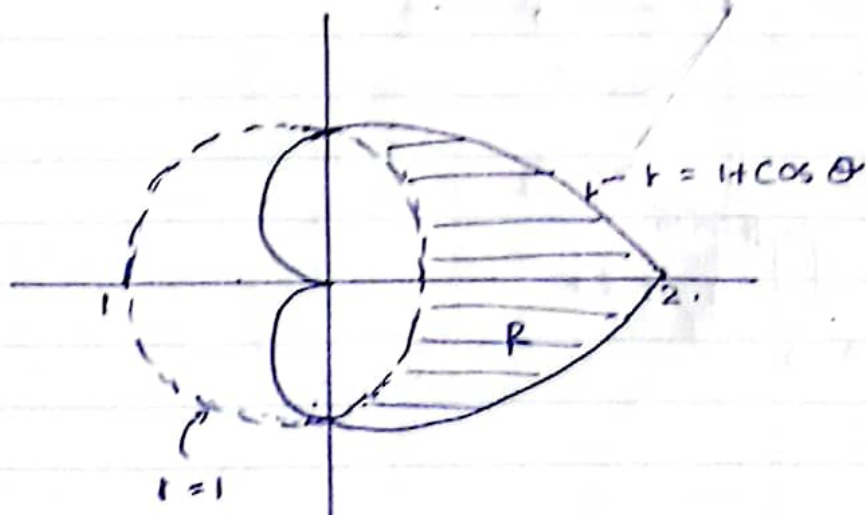
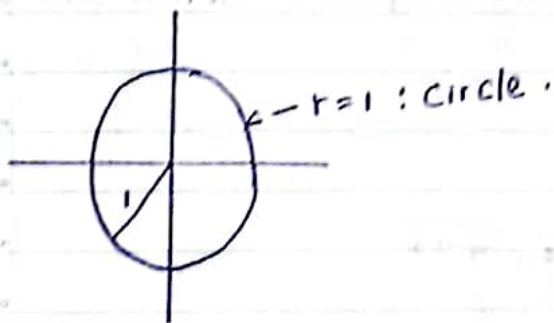
$$\begin{aligned}
 \iint_R e^{-(x^2+y^2)} dA &= \iint_D e^{-r^2} r dr d\theta \\
 &= \int_{\theta=0}^{2\pi} \int_{r=1}^2 r e^{-r^2} dr d\theta \\
 &= \left[\int_{\theta=0}^{2\pi} d\theta \right] \left[\int_{r=1}^2 r e^{-r^2} dr \right] \\
 &= \left[\theta \right]_{\theta=0}^{2\pi} \left[\frac{e^{-r^2}}{-2} \right]_{r=1}^2 \\
 &= [2\pi] \left[\frac{e^{-1} - e^{-4}}{2} \right] \\
 &= \pi \left(\frac{1}{e^4} - \frac{1}{e} \right)
 \end{aligned}$$

H.W :-

Find the area of the region R that lies inside the cardioid $r = 1 + \cos \theta$ and outside the circle $r = 1$



Cardioid: $r = a(1 - \cos \theta)$, $a > 0$,
 $0 \leq \theta \leq 2\pi$



The area is,

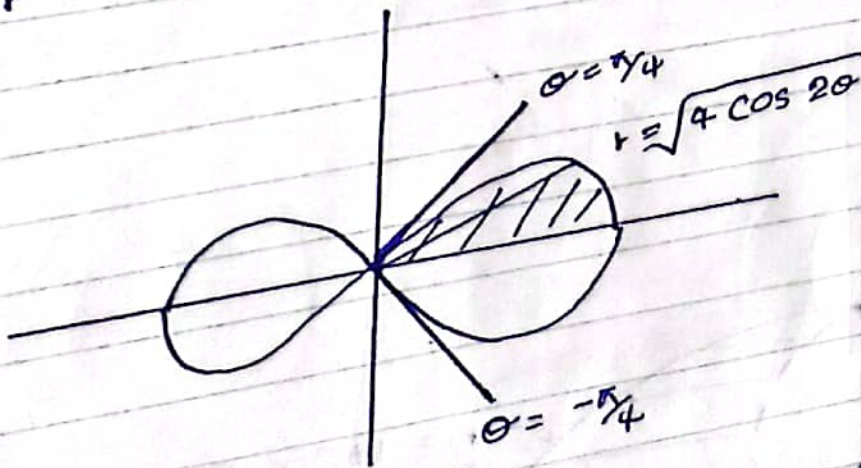
$$A(R) = \iint_R 1 \times r \, dr \, d\theta$$

$$= \int_{\theta = -\pi/2}^{\pi/2} \int_{r=1}^{1+\cos\theta} 1 \times r \, dr \, d\theta$$

$$= \int_{\theta = -\pi/2}^{\pi/2} \left[\frac{r^2}{2} \right]_{r=1}^{1+\cos\theta} d\theta$$

$$= \int_{\theta = -\pi/2}^{\pi/2} \frac{1}{2} \left[(1+\cos\theta)^2 - 1 \right] d\theta$$

Find the area enclosed by the lemniscate
 $r^2 = 4 \cos 2\theta$



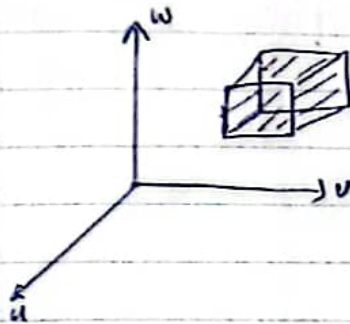
$$\text{Area} = \int_{\theta=0}^{\pi/4} \int_{r=0}^{\sqrt{4 \cos 2\theta}} 1 \, dr \, d\theta$$

Change of variables in triple Integrals:-

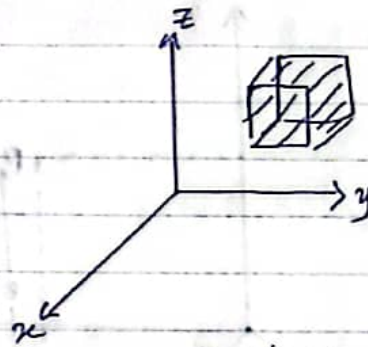
$$\text{Let } x = x(u, v, w)$$

$$y = y(u, v, w)$$

$$z = z(u, v, w)$$



Cartesian uvw-space



Cartesian xyz-space.

$$\frac{\partial(x, y, z)}{\partial(u, v, w)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} & \frac{\partial x}{\partial w} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} & \frac{\partial y}{\partial w} \\ \frac{\partial z}{\partial u} & \frac{\partial z}{\partial v} & \frac{\partial z}{\partial w} \end{vmatrix}$$

$$\iiint_V f(x, y, z) \, dv = \iiint_S f[x(u, v, w), y(u, v, w), z(u, v, w)] \frac{\partial(x, y, z)}{\partial(u, v, w)} \, du \, dv \, dw$$

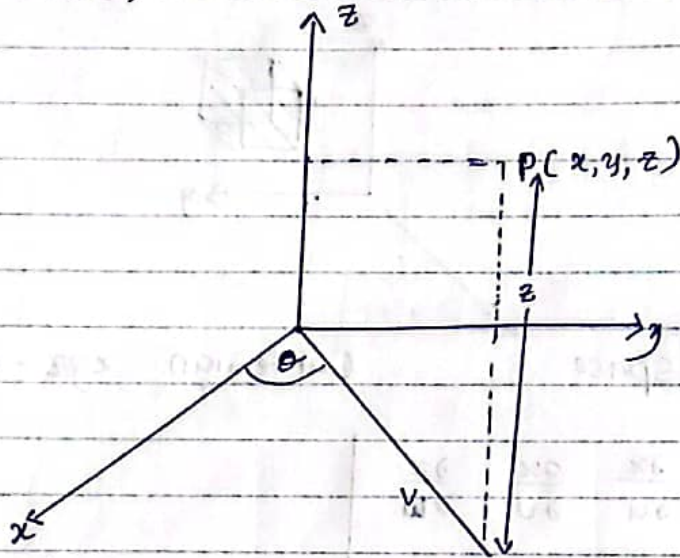
The above equation is valid under the conditions similar to those for the above integration.

Cylindrical Polar Coordinates :-

$$x = r \cos \theta$$

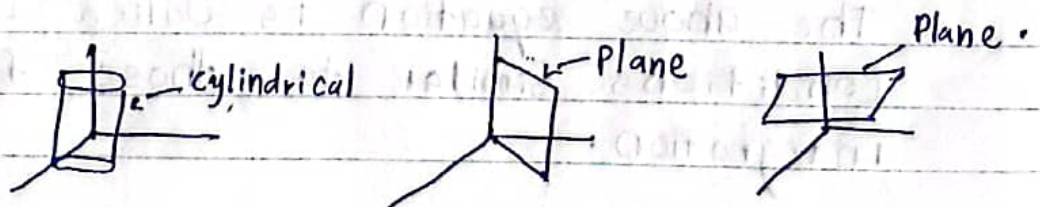
$$y = r \sin \theta$$

$$z = z, \text{ where } r > 0 \text{ and } 0 \leq \theta \leq 2\pi.$$



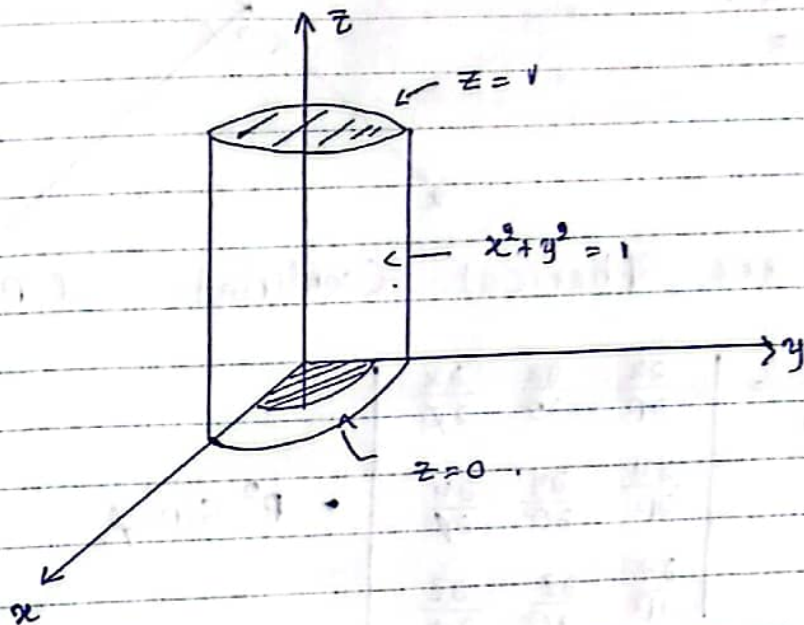
r , θ , and z are 'Cylindrical' coordinates of point P .

$$\frac{\partial(x, y, z)}{\partial(r, \theta, z)} = \begin{vmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} & \frac{\partial x}{\partial z} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} & \frac{\partial y}{\partial z} \\ \frac{\partial z}{\partial r} & \frac{\partial z}{\partial \theta} & \frac{\partial z}{\partial z} \end{vmatrix}$$



Example :-

Evaluate $\iiint_E (x^2 + y^2) z \, dx \, dy \, dz$, where E is the region enclosed by the cylinder $x^2 + y^2 = 1$ and the planes: $z = 0$ and $z = 1$. Use cylindrical coordinates.



using cylindrical coordinates, we have

$$\iiint_E (x^2 + y^2) z \, dx \, dy \, dz.$$

$$= \int_{z=0}^1 \int_{\theta=0}^{2\pi} \int_{r=0}^1 r^2 z \cdot r \, dr \, d\theta \, dz$$

$$= \left[\int_{z=0}^1 z \, dz \right] \left[\int_{\theta=0}^{2\pi} d\theta \right] \left[\int_{r=0}^1 r^3 \, dr \right]$$

$$= \left[\frac{z^2}{2} \right]_{z=0}^1 \left[\theta \right]_{\theta=0}^{2\pi} \left[\frac{r^4}{4} \right]_{r=0}^1$$

$$= \left(\frac{1}{2} \right) (2\pi) \frac{1}{4}$$

$$= \frac{\pi}{4}$$

this point.

Spherical polar Coordinates.

$$\rho = r > 0$$

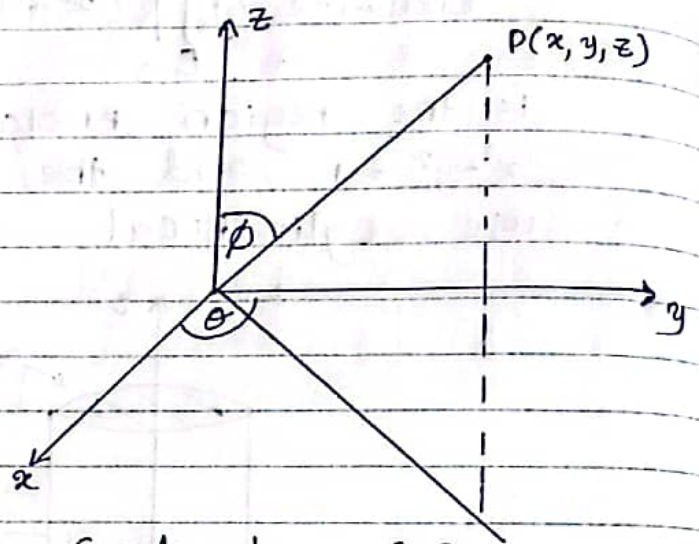
$$x = \rho \sin \theta \cos \phi$$

$$y = \rho \sin \theta \sin \phi$$

$$z = \rho \cos \theta$$

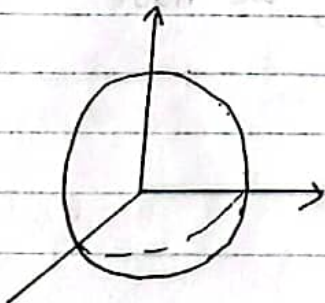
$$0 \leq \theta \leq 2\pi$$

$$0 \leq \phi \leq \pi$$

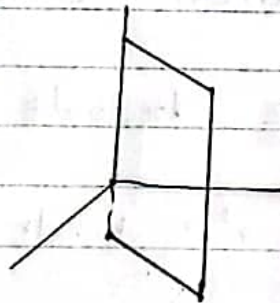


(ρ, ϕ, θ) are Spherical Coordinates of P .

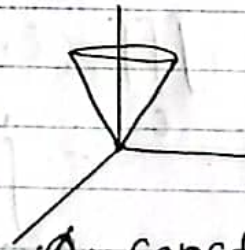
$$\frac{\partial(x, y, z)}{\partial(\rho, \phi, \theta)} = \begin{vmatrix} \frac{\partial x}{\partial \rho} & \frac{\partial x}{\partial \phi} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial \rho} & \frac{\partial y}{\partial \phi} & \frac{\partial y}{\partial \theta} \\ \frac{\partial z}{\partial \rho} & \frac{\partial z}{\partial \phi} & \frac{\partial z}{\partial \theta} \end{vmatrix} = \rho^2 \sin \phi$$



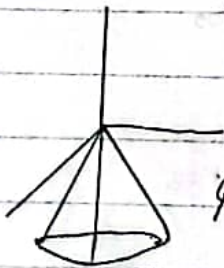
$\rho = \text{Constant}$
Sphere.



$\theta = \text{Constant}$
half plane



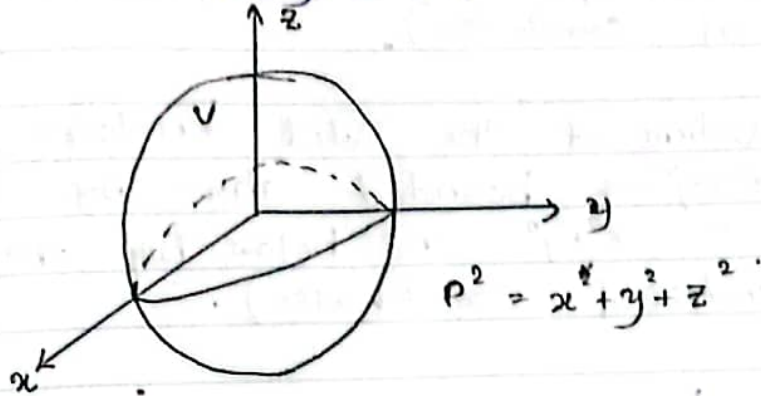
$\phi = \text{Constant} = c$
 $0 \leq c \leq \frac{\pi}{2}$
Half



$\phi = \text{Constant} = c$
 $\frac{\pi}{2} < c < \pi$

Example :-

Find $\iiint_V \sqrt{x^2+y^2+z^2} \, du$, where u is given by $x^2+y^2+z^2 \leq 1$.



$$\iiint_V \sqrt{x^2+y^2+z^2} \, du = \int_{\phi=0}^{\pi} \int_{\theta=0}^{2\pi} \int_{\rho=0}^1 \rho \cdot \rho^2 \sin \phi \, d\phi \, d\theta \, d\rho.$$

$$= \left[\int_{\phi=0}^{\pi} \sin \phi \, d\phi \right] \left[\int_{\theta=0}^{2\pi} d\theta \right] \left[\int_{\rho=0}^1 \rho^3 \, d\rho \right]$$

$$= \left[-\cos \phi \right]_{\phi=0}^{\pi} \left[\theta \right]_{\theta=0}^{2\pi} \left[\frac{\rho^4}{4} \right]_{\rho=0}^1$$

$$= -(-1-1)(2\pi) \cdot \frac{1}{4}$$

$$= \pi$$

H.W :-

- 01) Find the Volume of the upper region D cut from the Solid sphere $\rho \leq 1$ by the Cone $\phi = \frac{\pi}{4}$. (use spherical coordinates).
- 02) Find the Volume of the Solid enclosed by the Cylinder $x^2 + y^2 = 4$ bounded above by the parabolic $z = x^2 + y^2$ and below by the xy -plane (use Cylindrical coordinates).

